

5G RAN

Massive MIMO iBeam (Low-Frequency TDD) Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

None

- Editorial changes

Revised the descriptions of the principles of robust weights and self-fusion weights. For details, see [5.1.1 Weight Enhancement](#).

Revised the descriptions of impacts in [3.2.2 Impacts](#), [4.2.2 Impacts](#), and [5.2.2 Impacts](#).

Revised the description of the impact relationship between precise RB control for downlink retransmissions and power optimization for downlink retransmissions. For details, see [5.3.2.3 Function Impacts](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 05 (2025-09-26).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added iBeam 3.0. For details, see: • 2.2 Features in This Document • 2.3 Differences • 5 iBeam 3.0	• Added parameters: – NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw – NRDUCellPdschPrerecode.SelfFusionWtSrsPresinrThld – NRDUCellPdschPrerecode.SelfFusionWtRhoThld – NRDUCellPdcchAlgo.PdcchRobWtSrsPresinrThld – NRDUCellPdcchAlgo.PdcchRobWtBFGainCoeff • Modified parameters: – Added the DL_SELF_FUSION_WEIGHT_SW option to the NRDUCellPdschPrerecode.PrecodingAlgoSwitch parameter. – Added the PDCCH_ROBUST_WEIGHT_SW option to the NRDUCellPdcch.PdcchAlgoSwitch parameter. – Added the DL_SMART_AMC_SW option to the NRDUCellDLMimo.DLMuMimoSchSupplementSw parameter. – Added the DL_MU_MCS_INTEL_OPT_SW option to the NRDUCellAiAlgo.AiAmcAlgoSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
	- Added the ROBUST_RANK_SW and ROBUST_RANK_MOBILITY_OPT_SW options to the NRDUCellDlAmc.DlMcsSelAlgoSw parameter. - Added the DL_SMALL_PACKET_SHAPE_SCH_SW option to the NRDUCellDlSch.DlSchAlgoSwitch parameter. - Added the DL_RETRANS_PRR_OPT_SW option to the NRDUCellChnPwr.ChnPwrAlgoSwitch parameter. - Added the SRS_ENH_BLIND_IS_SW option to the NRDUCellSrs.SrsDetectionAlgoSwitch parameter. - Added the DL_RETRANS_PRI_ADJ_SW option to the NRDUCellDlSch.DlSchAlgoSwitch parameter. - Added the DL_SLOT_AWARE_SCH_OPT_SW option to the NRDUCellDlSch.DlSchAlgoSwitch parameter.		

Change Description	Parameter Change	RAT	Base Station Model
Added impact relationships between L/NR Inter-band Deep Coordination and the following functions: PDCCH aggregation level reduction and CCE partial symbol adaptation. For details, see: • 3.3.2.3 Function Impacts • 4.3.2.3 Function Impacts	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between adaptive filtering for SRS channel estimation and PDSCH self-fusion weights. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes

Revised the descriptions of cell requirements. For details, see [3.3.5 Cells](#) and [4.3.4 Cells](#).

Revised the descriptions of the principles of downlink multi-BWP interference avoidance. For details, see [Downlink Multi-BWP Interference Avoidance](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-081201	iBeam	3 iBeam
FOFD-091200	iBeam 2.0	4 iBeam 2.0
FOFD-100200	iBeam 3.0	5 iBeam 3.0

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
iBeam	None	3 iBeam
iBeam 2.0	None	4 iBeam 2.0
iBeam 3.0	None	5 iBeam 3.0

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
iBeam	Supported only in low frequency bands	3 iBeam
iBeam 2.0	Supported only in low frequency bands	4 iBeam 2.0
iBeam 3.0	Supported only in low frequency bands	5 iBeam 3.0

3 iBeam

3.1 Principles

The rapid growth in the number of 5G UEs increases cell loads and, when multiple cells offer contiguous coverage, also increases interference. Scheduling resources are prone to become insufficient, and UE performance in cells consequently deteriorates. Intelligent Beam (iBeam) is designed to address this issue. This feature applies to TDD massive MIMO cells. It improves downlink performance and increases the downlink user-perceived rate through interference suppression enhancement, scheduling enhancement, and other algorithms.

To enable iBeam and its various functions for interference suppression enhancement and scheduling enhancement, the **NRDUCellFeatureSw.HighPrecisionBeamSwitch** parameter must be set to **ON**.

3.1.1 Interference Suppression Enhancement

Blind Interference Suppression for SRS Measurement

A gNodeB can estimate uplink channel characteristics based on sounding reference signal (SRS). In addition, thanks to reciprocity between uplink and downlink channels, a gNodeB can deduce downlink channel characteristics from uplink channel characteristics. On this account, channel estimation depends on the accuracy and effectiveness of SRS measurement, which is easily affected by SRS interference.

The function of blind interference suppression for SRS measurement uses the differences in the characteristics of interference signals and target signals to identify interference signals more effectively. This helps suppress SRS interference to improve SRS measurement accuracy and increase the downlink user-perceived rate.

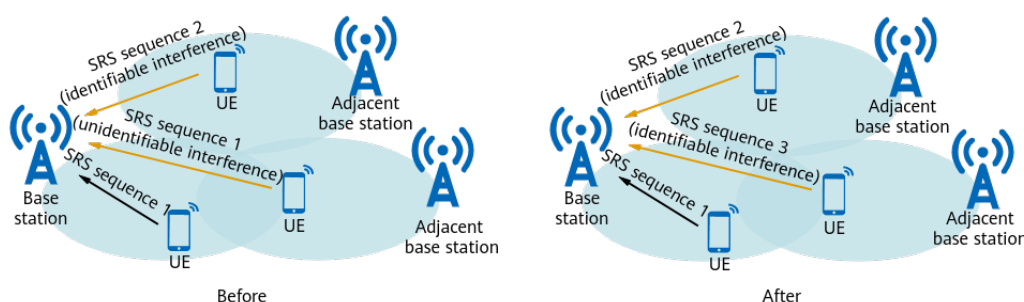
Blind interference suppression for SRS measurement is controlled by the **SRS_BLIND_IS_MEAS_SW** option of the **NRDUCellSrsMeas.SrsMeasOptSwitch** parameter.

Tight SRS Multiplexing

SRS sequences are generated based on Zadoff-Chu (ZC) sequences. After obtaining information about SRS sequences and resources configured on gNodeBs, UEs will generate SRSs. Adjacent cells offering contiguous coverage may use the same SRS sequence. In this case, gNodeBs cannot separate SRSs sent by UEs in different adjacent cells. This causes SRS measurement errors and decreases the downlink user-perceived rate.

The tight SRS multiplexing function staggers the SRS sequences used in different cells to reduce the probability of SRS sequence conflicts. This function improves SRS measurement accuracy and increases the downlink user-perceived rate. [Figure 3-1](#) illustrates this function.

Figure 3-1 Tight SRS multiplexing



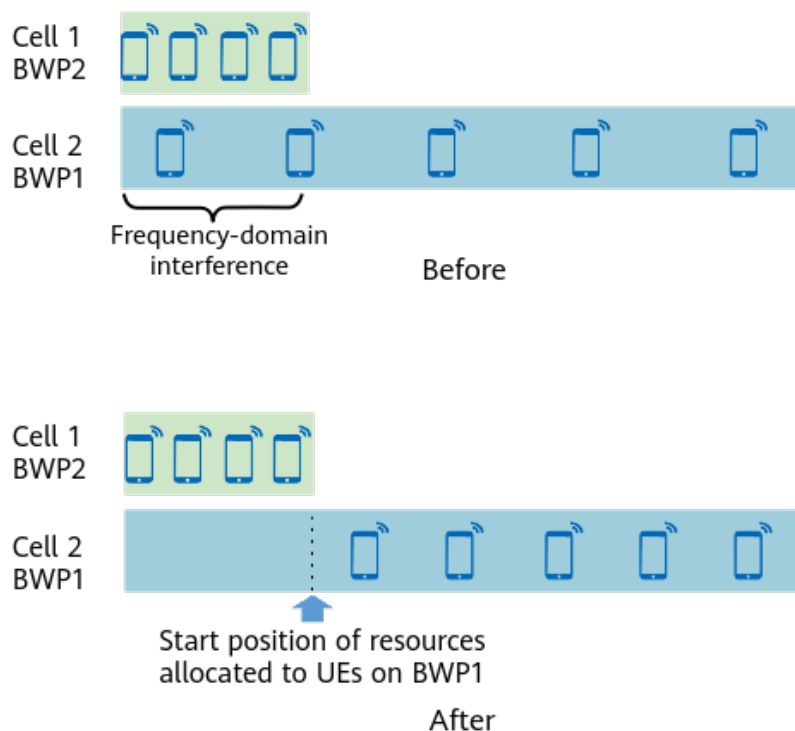
Tight SRS multiplexing is controlled by the **SRS_TIGHT_MULTIPLEXING_SW** option of the **NRDUCellSrs.SrsAlgoSwitch** parameter.

Downlink Multi-BWP Interference Avoidance

If the power saving BWP function is enabled and most UEs work on BWP2, there is intra-frequency interference between BWP2 in one cell and BWP1 in another cell. (BWP is short for bandwidth part.) For details about power saving BWP, see *UE Power Saving*.

Downlink multi-BWP interference avoidance can mitigate such interference. As illustrated in [Figure 3-2](#), with this function, the gNodeB preferentially allocates frequency-domain resources on the spectrum unoccupied by BWP2 in cell 1 to UEs working on the full-bandwidth BWP1 in cell 2. This eases the interference between BWP1 in cell 2 and BWP2 in cell 1 and thereby increases the downlink user-perceived rate. If UEs working on BWP1 have occupied all RBs on the non-BWP2 spectrum and still have data to be scheduled, these UEs can use the remaining RBs, if available, on BWP2.

Figure 3-2 Downlink multi-BWP interference avoidance



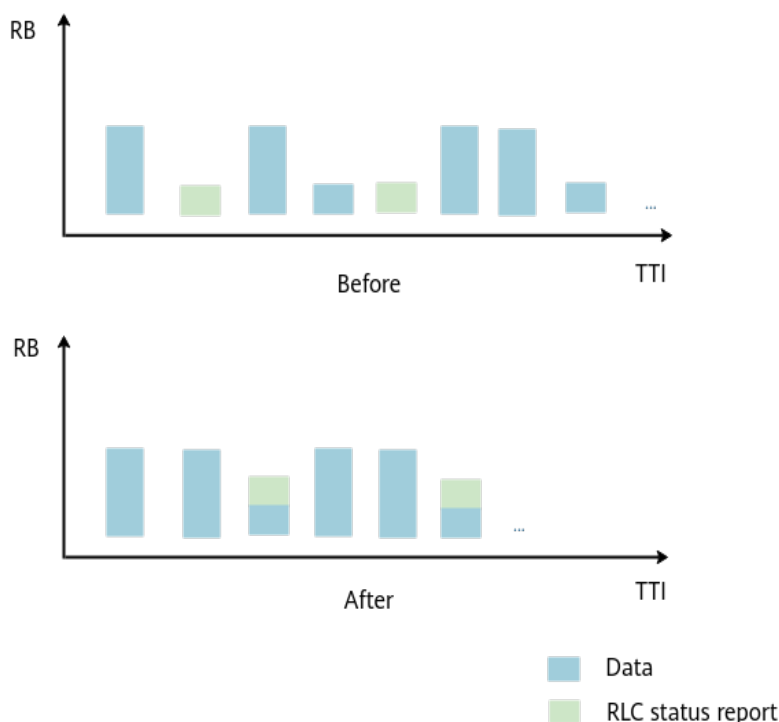
Downlink multi-BWP interference avoidance is controlled by the **DL_BWP_HYBRID_INTRF_RANDOM_SW** option of the **NRDUCellDISch.DISchAlgoSwitch** parameter.

3.1.2 Scheduling Enhancement

Merged Scheduling for Downlink RLC Status Reports

When acknowledged mode (AM) is used at the Radio Link Control (RLC) layer, RLC status reports of uplink data need to be scheduled in the downlink. Separate transmission of RLC status reports consumes control channel elements (CCEs). This may affect user experience when PDCCH CCEs are insufficient or the downlink PRB usage is high. For less CCE resource consumption and better user experience, merged scheduling for downlink RLC status reports can be used. This function postpones the transmission of RLC status reports, which used to be separately transmitted, and then schedules these reports together with downlink data, as illustrated in [Figure 3-3](#).

Figure 3-3 Merged scheduling for downlink RLC status reports



Merged scheduling for downlink RLC status reports is controlled by the **DL_RLC_STAT_RPT_MERGE_SCH_SW** option of the **NRDUCellDISch.DISchAlgoSwitch** parameter.

PDCCH Aggregation Level Reduction

A request for PDCCH resources with a high aggregation level may lead to a resource allocation failure, which affects user experience. With the function of PDCCH aggregation level reduction, if PDCCH CCEs fail to be allocated to a UE, another attempt at CCE allocation will be made for the UE with the CCE aggregation level further reduced and the CCE power spectral density (PSD) increased.

PDCCH aggregation level reduction is controlled by the **PDCCH_AGG_LVL_COMPR_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter. This function takes effect only when the PDCCH CCE usage (equal to the result of **N.CCE.Used.Avg/N.CCE.Avail.Avg**) is less than the threshold specified by **NRDUCellPdcchAlgo.AggLvlComprCceUsageThld**.

Beam Selection Optimization

On live networks, when the **NRDUCellBeamAlgo.BeamPerceiveMode** parameter is set to **DISTRIBUTED_MODE** to enable MetaAAU beam sensing, the beam accuracy is adaptively enhanced. Beam selection optimization further enhances the beam accuracy based on the UE traffic model for higher downlink UE throughput in cells.

Beam selection optimization is controlled by the **BEAM_SELECT_OPT_SW** option of the **NRDUCellBeamAlgo.ChannelOptAlgoSwitch** parameter.

Downlink MU Precise Scheduling

Resources may be wasted when RBs with different lengths are paired. To improve resource utilization, downlink MU precise scheduling is introduced. This function adjusts resource allocation for MU-MIMO UEs to increase their spectral efficiency, which leads to higher downlink UE throughput in cells.

Downlink MU precise scheduling is controlled by the **DL_MU_PRECISE_SCH_SW** option of the **NRDUCellDIMimo.DLMuMimoSchSupplementSw** parameter.

Downlink MU Anti-Interference Scheduling

In MU scheduling, higher ranks of UEs in groups result in more layers but stronger interference between groups. Downlink MU anti-interference scheduling reduces both the ranks and power of to-be-scheduled UEs to mitigate the interference between paired UEs and to neighboring cells. This increases the downlink user-perceived rate.

Downlink MU anti-interference scheduling is controlled by the **DL_MU_ANTI_INTRF_SCH_SW** option of the **NRDUCellDIMimo.DLMuMimoSchSupplementSw** parameter. When this option is selected, it is recommended that MCS optimization for tail packets also be enabled for greater benefits. (MCS optimization for tail packets is controlled by the **TAIL_PKT_MCS_OPT_SW** option of the **NRDUCellDISch.DISchAlgoSwitch** parameter.)

The downlink MU-MIMO grouping policy is specified by the **NRDUCellPdsch.DLMuMimoGroupMode** parameter. This parameter can be set to **BEAM_GAP**, **ISOLATION_CORRELATION**, **BEAM_ENHANCE**, or **CLUSTER_GROUP** for downlink MU anti-interference scheduling.

Resource-based Adaptive Downlink Scheduling

Resource-based adaptive downlink scheduling adjusts scheduling priorities in real time based on the RBs occupied by UEs and the traffic model. The aim is to increase the downlink user-perceived rate by allocating contiguous resources to UEs whenever possible with the shortest total transmission delay.

Resource-based adaptive downlink scheduling is controlled by the **RES_BASED_DL_ADAPT_SCH_SW** option of the **NRDUCellDISch.DISchAlgoSwitch** parameter.

3.2 Network Analysis

3.2.1 Benefits

iBeam increases the **User Downlink Average Throughput (DU)** and downlink average MCS index (equal to $\text{SUM}(k \times \text{N.ChMeas.PDSCH.MCS}.k) / \text{SUM}(\text{N.ChMeas.PDSCH.MCS}.k)$, where k ranges from 0 to 28). The actual benefits of iBeam depend on factors such as the traffic model, network load, and physical cell identifier (PCI) planning. The higher the downlink PRB usage and the stronger the SRS interference from neighboring cells, the greater the benefits.

iBeam is recommended in intra-frequency contiguous networking where all the following conditions are met:

- The downlink PRB usage (equal to $\text{N.PRB.DL.Used.Avg}/\text{N.PRB.DL.Avail.Avg}$) ranges from 30% to 50%.
- The SRS interference (indicated by N.SRS.NI.Avg) from neighboring cells is greater than -105 dBm.
- The number of RRC_CONNECTED UEs in all cells exceeds 300 on a base station equipped with the UBBPfw or exceeds 100 on a base station equipped with other baseband processing units. The number of RRC_CONNECTED UEs in a cell is indicated by $\text{N.User.RRCConn.Avg}$.
- The proportion of PRBs paired for downlink MU-MIMO is greater than 10%. This proportion equals $\text{N.ChMeas.MIMO.DL.Pair.PRb}/(\text{N.PRB.DL.Used.Avg} \times \text{N.DL.PDSCH.Tti.Num}) \times 100\%$.

If the preceding conditions are not met, some functions offer fewer or even no benefits. iBeam will offer fewer benefits if any of its function for interference suppression enhancement or scheduling enhancement is disabled. In addition, it is recommended that tight SRS multiplexing be enabled when multiple cells offer contiguous coverage in intra-frequency networking. If it is enabled in other scenarios, the function will offer fewer benefits.

When downlink MU precise scheduling is enabled, it is recommended that MCS optimization for tail packets also be enabled for greater benefits. (MCS optimization for tail packets is controlled by the **TAIL_PKT_MCS_OPT_SW** option of the **NRDUCellDisch.DLSchAlgoSwitch** parameter.)

3.2.2 Impacts

The impacts between this function and other functions are described in [3.3.2.3 Function Impacts](#).

iBeam may have the following impacts on the network:

- The **User Uplink Average Throughput (DU)** may change.
- The service drop rate in a cell may change. (Service drop rate in a cell = $\text{N.UECntx.AbnormRel}/(\text{N.UECntx.NormRel} + \text{N.UECntx.AbnormRel})$)
- The downlink PRB usage may decrease. (Downlink PRB usage = $\text{N.PRB.DL.Used.Avg}/\text{N.PRB.DL.Avail.Avg}$)
- The RRC connection reestablishment rate may decrease. (RRC connection reestablishment rate = $\text{N.RRC.ReEst.Att}/(\text{N.RRC.SetupReq.Msg} + \text{N.RRC.ReEst.Att})$)
- The MU-MIMO pairing rate may change. (MU-MIMO pairing rate = $\text{N.ChMeas.MIMO.DL.Pair.PRb}/(\text{N.PRB.DL.Used.Avg} \times \text{N.DL.PDSCH.Tti.Num})$)
- The average number of layers paired for downlink MIMO (indicated by the **N.ChMeas.MIMO.DL.Pair.Layer.Avg** counter) may decrease.
- The average SRS interference (indicated by the **N.SRS.NI.Avg** counter) may change.
- The values of the following scheduling-related indicators may change:
 - Downlink average rank, which equals $(1 \times \text{N.PDSCH.InitTbDL.Rank1} + 2 \times \text{N.PDSCH.InitTbDL.Rank2} + 3 \times \text{N.PDSCH.InitTbDL.Rank3} + 4 \times \text{N.PDSCH.InitTbDL.Rank4})/(\text{N.PDSCH.InitTbDL.Rank1} +$

- N.PDSCH.InitTbDL.Rank2 + N.PDSCH.InitTbDL.Rank3 + N.PDSCH.InitTbDL.Rank4)**
- Uplink and downlink IBLERs
 - Uplink IBLER = $(N.UL.SCH.HalfPiBPSK.ErrTB.Ibler + N.UL.SCH.QPSK.ErrTB.Ibler + N.UL.SCH.16QAM.ErrTB.Ibler + N.UL.SCH.64QAM.ErrTB.Ibler) / (N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)$
 - Downlink IBLER = $(N.DL.SCH.QPSK.ErrTB.Ibler + N.DL.SCH.16QAM.ErrTB.Ibler + N.DL.SCH.64QAM.ErrTB.Ibler + N.DL.SCH.256QAM.ErrTB.Ibler) / (N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB + N.DL.SCH.1024QAM.TB)$
 - Uplink and downlink RBLERs
 - Uplink RBLER = $(N.UL.SCH.HalfPiBPSK.ErrTB.Rbler + N.UL.SCH.QPSK.ErrTB.Rbler + N.UL.SCH.16QAM.ErrTB.Rbler + N.UL.SCH.64QAM.ErrTB.Rbler) / (N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)$
 - Downlink RBLER = $(N.DL.SCH.QPSK.ErrTB.Rbler + N.DL.SCH.16QAM.ErrTB.Rbler + N.DL.SCH.64QAM.ErrTB.Rbler + N.DL.SCH.256QAM.ErrTB.Rbler) / (N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB + N.DL.SCH.1024QAM.TB)$
 - The proportion of samples in which the average downlink UE throughput in a cell falls within low-throughput ranges may change. (This proportion = $(N.Thp.DL.Samp.Index0 + N.Thp.DL.Samp.Index1) / \text{SUM}(N.Thp.DL.Samp.Indexk)$, where k is in the range of 0 to 14)
 - The total duration of UEs working on BWP2 in a cell (indicated by the **N.Bwp2.Dur.Total** counter) may change.
 - The proportion of traffic volumes on BWP2 may change. (Proportion of traffic volumes on BWP2 = $(N.ThpVol.DL - N.BWP.ThpVol.DL) / N.ThpVol.DL$)
 - The CCE allocation success rate in a cell may increase. (CCE allocation success rate in a cell = $(N.CCE.DL.AggLvl1Num + N.CCE.DL.AggLvl2Num + N.CCE.DL.AggLvl4Num + N.CCE.DL.AggLvl8Num + N.CCE.DL.AggLvl16Num) / N.CCE.DL.AllocReq.Num$)
 - Enabling or disabling tight SRS multiplexing for a cell will cause the cell to restart and all UEs in the cell to reaccess the network.

3.3 Requirements

3.3.1 Licenses

Feature ID	Feature Name	Model	NE	Sales Unit
FOFD-081201	iBeam	NR0S00BEA M00	gNodeB	Per Cell
For the license control item NR0S00BEAM00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the NRDUCellFeatureSw.HighPrecisionBeamSwitch parameter cannot be modified due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.				

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

3.3.2.1 Prerequisite Functions

- Blind interference suppression for SRS measurement

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>Channel Management</i>	Blind interference suppression for SRS measurement can be enabled only after UE-characteristic-based SRS period adaptation is enabled.
Low-frequency TDD	High-precision beam	NRDUCellFeatureSw.HighPrecisionBeamSwitch	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Blind interference suppression for SRS measurement can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamSwitch parameter is set to ON .

- Tight SRS multiplexing

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS interference coordination based on self-contained slots	SRS_INTRF_COORD_S_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Channel Management</i>	For low-speed cells (with NRDUCell.HighSpeedFlag set to LOW_SPEED), tight SRS multiplexing can be enabled only when SRS interference coordination based on self-contained slots is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS period and CDM adaptation	SRS_PERIOD_AND_CDM_ADAPT_SW option of the NRDUCellHighSpeedMob.HighSpeedAlgoOptSw parameter	<i>High Speed Mobility</i>	For high-speed cells (with NRDUCell.HighSpeedFlag set to HIGH_SPEED), tight SRS multiplexing can be enabled only when SRS period and code division multiplexing (CDM) adaptation is enabled.
Low-frequency TDD	High-precision beam	NRDUCellFeatureSw.HighPrecisionBeamSwitch	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Tight SRS multiplexing can be enabled only when the NRDUCellFeatureSw.HighPrecisionBeamSwitch parameter is set to ON .

- Downlink multi-BWP interference avoidance

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving BWP	BWP2_SWITCH option of the NRDUCellUePowerSaving.BwpPowerSavingSw parameter	<i>UE Power Saving</i>	Downlink multi-BWP interference avoidance can be enabled only when power saving BWP is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-precision beam	NRDUCellFeatureSw.HighPrecisionBeamSwitch	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Downlink multi-BWP interference avoidance can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamSwitch parameter is set to ON .

- PDCCH aggregation level reduction

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Maximum Aggregated PDCCH Power Offset	NrDuCellChnPower.MaxPdcchAggPwrOffset	None	PDCCH aggregation level reduction can be enabled only when Maximum Aggregated PDCCH Power Offset is set a value greater than or equal to 60.

- Beam selection optimization

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	MetaAAU beam sensing	NRDUCellBeamAlgo.BeamPerceiveMode set to DISTRIBUTED_MODE	<i>MIMO (TDD)</i>	Beam selection optimization can take effect only when MetaAAU beam sensing is in effect and the NRDUCellFeatureSw.HighPrecisionBeamSwitch parameter is set to ON .
Low-frequency TDD	High-precision beam	NRDUCellFeatureSw.HighPrecisionBeamSwitch	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	

- Downlink MU precise scheduling

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Downlink MU precise scheduling can take effect only when downlink MU-MIMO is in effect and the NRDUCellFeatureSw.HighPrecisionBeamSwitch parameter is set to ON .
Low-frequency TDD	High-precision beam	NRDUCellFeatureSw.HighPrecisionBeamSwitch	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	

- Downlink MU anti-interference scheduling

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Downlink MU anti-interference scheduling can take effect only when downlink MU-MIMO is in effect and the NRDUCellFeatureSw.HighPrecisionBeamSwitch parameter is set to ON .
Low-frequency TDD	High-precision beam	NRDUCellFeatureSw.HighPrecisionBeamSwitch	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	
Low-frequency TDD	Spectral-efficiency-based adaptive switching between SU-MIMO and MU-MIMO	DL_SE_SU_MU_MIMO_ADAPT_SW option of the NRDUCellDimmo.DLMuMimoSchSupplementSw parameter	<i>MIMO (TDD)</i>	Downlink MU anti-interference scheduling can take effect only when spectral-efficiency-based adaptive switching between SU-MIMO and MU-MIMO is in effect.

- Resource-based adaptive downlink scheduling

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Resource-based adaptive scheduling for massive MIMO	RES_BASED_MM_ADAPT_SCH_SW option of the NRDUCellServExp.ServiceExpAlgoSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	Resource-based adaptive downlink scheduling can take effect only when resource-based adaptive scheduling for massive MIMO is in effect and the NRDUCellFeatureSw.HighPrecisionBeamSwitch parameter is set to ON .
Low-frequency TDD	High-precision beam	NRDUCellFeatureSw.HighPrecisionBeamSwitch	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	

3.3.2.2 Mutually Exclusive Functions

- iBeam

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced muting policy	NRDUCell.ScenarioPolicy set to CELL_ENH_MUTING_POLICY	None	iBeam does not work with the enhanced muting policy.

- Blind interference suppression for SRS measurement

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	None

- Tight SRS multiplexing

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low-speed UE performance optimization	LOW_SPEED_PERFORMANCE_OPT_SW option of the NRCellAlgoSwitch.HighSpeedDedPolicySw parameter	<i>High Speed Mobility</i>	For high-speed cells (with NRDUCell.HighSpeedFlag set to HIGH_SPEED), if tight SRS multiplexing is enabled, low-speed UE performance optimization must be disabled.

- Downlink multi-BWP interference avoidance

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE TDD and NR spectrum sharing	LTE_NR_TDD_SPECT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	None	None

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	External interference avoidance	• NRDUCellIntrfIdent.IntrfIdentificationMethod set to UL_DL_MANUAL • UL_CELL_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter • DL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter	<i>Interference Avoidance</i>	Downlink multi-BWP interference avoidance must be disabled when all the following conditions are met: • Interference ranges are manually configured for both uplink and downlink interference identification. • Cell-level interference avoidance is used in the uplink. • UE-level interference avoidance is used in the downlink.
Low-frequency TDD	Bi-Carrier	BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	None	None

3.3.2.3 Function Impacts

- Tight SRS multiplexing

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	Enabling tight SRS multiplexing may decrease the average uplink throughput of UEs working on the power saving BWP.

- Downlink multi-BWP interference avoidance

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multiple BWP1s for RedCap	NRDUCellRedCap.RedCapUeMaxAvailBandwidth not set to 20M	<i>RedCap</i>	When both downlink multi-BWP interference avoidance and multiple BWP1s for RedCap are enabled and multiple RedCap UEs access the network through different BWP1s, the downlink interference on each BWP1 may change. As a result, the performance of RedCap UEs may be affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink interference randomization-based scheduling	DL_INTRF_RAN DOM_SW option of the NRDUCellAlgo Switch.AdaptiveEdgeExpEnhSwitch parameter	<i>Downlink Scheduling</i>	If downlink multi-BWP interference avoidance is not in effect, resources on the entire cell spectrum can be used to randomize interference on BWP1 between cells. If downlink multi-BWP interference avoidance is in effect, only resources on the non-BWP2 spectrum can be used for the same purpose, which may reduce the benefits provided by downlink interference randomization-based scheduling.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	When the function of low latency and high reliability is enabled together with power saving BWP, UEs running low-latency and high-reliability services will not work on BWP2. In addition, as low-latency and high-reliability services have a higher priority, RBs are reserved for these services. As a result, downlink multi-BWP interference avoidance provides fewer benefits in this scenario.
Low-frequency TDD	Start RB avoidance	START_RB_AVOID_SW option of the NRDUCellRedCap.RedCapAlgorithmSwitch parameter	<i>RedCap</i>	Downlink multi-BWP interference avoidance does not take effect when enabled together with start RB avoidance.

- Merged scheduling for downlink RLC status reports

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Symbol power saving	NRDUCellAlgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	When both merged scheduling for downlink RLC status reports and symbol power saving are in effect, the symbol power saving duration (measured by N.PowerSaving.SymbolShutdown.Dur) may increase.

- PDCCH aggregation level reduction

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	Only NR inter-carrier dynamic power sharing takes effect when it is enabled together with PDCCH aggregation level reduction.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	- LTE: LTE_NR_DYN_POWER_SHARING_SWITCH option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter - NR: LTE_NR_DYN_POWER_SHARING_SWITCH option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	Only LTE TDD and NR Flash Dynamic Power Sharing takes effect when it is enabled together with PDCCH aggregation level reduction.
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	Only NR multi-band power adaptive scheduling takes effect when it is enabled together with PDCCH aggregation level reduction.
Low-frequency TDD	LTE and NR multi-band power adaptive scheduling	LNR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	Only LTE and NR multi-band power adaptive scheduling takes effect when it is enabled together with PDCCH aggregation level reduction.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-BWP power sharing	INTER_BWP_POWER_SHARING_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter	<i>RedCap</i>	When PDCCH aggregation level reduction and inter-BWP power sharing are both enabled, the overall benefits provided by these two functions decrease.
Low-frequency TDD	RedCap power aggregation	REDCAP_POWER_AGGREGATION_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter	<i>RedCap</i>	When PDCCH aggregation level reduction and RedCap power aggregation are both enabled, the overall benefits provided by these two functions decrease.
Low-frequency TDD	L/NR Inter-band Deep Coordination	LNR_INTER_BAND_DEEP_COORD_SW option of the NRDUCellResSbrCfg.DynPwrShrAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	Only L/NR Inter-band Deep Coordination takes effect when it is enabled together with PDCCH aggregation level reduction.

- Downlink MU anti-interference scheduling

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CBF	INTRA_GNB_DL_CS_CBF_SW and INTER_GNB_DL_CS_CBF_SW options of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	Downlink MU anti-interference scheduling does not take effect in slots where coordinated beamforming (CBF) is in effect.
Low-frequency TDD	Fast CBF	QUICK_CBF_SW option of the NRDUCellComp.CsCbfAlgoSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	Downlink MU anti-interference scheduling does not take effect in slots where fast CBF is in effect.
Low-frequency TDD	DL MU-MIMO Group Mode	NRDUCellPdsch.DIMuMimoGroupMode set to ONE_USER_ON_E_GROUP	None	Downlink MU anti-interference scheduling does not take effect if the NRDUCellPdsch.DIMuMimoGroupMode parameter is set to ONE_USER_ON_E_GROUP .
Low-frequency TDD	DL SRS MU-MIMO Rank Mode	NRDUCellDLRank.DLSrsMuMimoRankMode set to FIX_MU_RANK	None	Downlink MU anti-interference scheduling does not take effect if the NRDUCellDLRank.DLSrsMuMimoRankMode parameter is set to FIX_MU_RANK .

- Resource-based adaptive downlink scheduling

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Resource-based adaptive scheduling for massive MIMO	RES_BASED_MM_ADAPT_SCH_SW option of the NRDUCellServExp.ServiceExpAlgoSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	Enabling resource-based adaptive scheduling for massive MIMO increases the DU delay and has other impacts. These impacts may become greater if resource-based adaptive downlink scheduling is also enabled.
Low-frequency TDD	Experience-based adaptive scheduling for massive MIMO	EXP_BASED_MM_ADAPT_SCH_SW option of the NRDUCellServExp.ServiceExpAlgoSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	When both resource-based adaptive downlink scheduling and experience-based adaptive scheduling for massive MIMO are enabled, the first packet delay (indicated by the N.Traffic.DL.RLCFirstPktDelay.Time counter) may increase.

3.3.3 Hardware

Base Station Models

- PDCCH aggregation level reduction, downlink multi-BWP interference avoidance, and beam selection optimization are supported by 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- Blind interference suppression for SRS measurement, tight SRS multiplexing, merged scheduling for downlink RLC status reports, downlink MU precise scheduling, downlink MU anti-interference scheduling, and resource-based adaptive downlink scheduling are supported by the following base station models:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- Beam selection optimization is supported by NR TDD-capable low-frequency UBBPg and later series baseband processing units and all NR TDD-capable low-frequency main control boards. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- The other functions are supported by all NR TDD-capable low-frequency main control boards and baseband processing units. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

For LampSite base stations, the RF module requirements are the same as those of distributed massive MIMO. For details about distributed massive MIMO, see *Distributed Antenna Solutions (Low-Frequency TDD)*.

For macro base stations:

- On non-distributed base stations, the compatible RF modules vary by functions:
 - PDCCH aggregation level reduction is only supported by the AAU563x and AAU533x series.
 - Beam selection optimization is only supported by MetaAAUs. For details about MetaAAU modules, see *Beam Management*.
 - The other functions are supported by all NR TDD-capable low-frequency RF modules with at least 32T32R. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- On distributed base stations, the RF modules must meet both of the following requirements:
 - The RF module requirements for distributed massive MIMO on macro base stations must be met. For details about distributed massive MIMO, see *Distributed Antenna Solutions (Low-Frequency TDD)*.
 - The RF modules must support distributed massive MIMO cells with at least 32T.

3.3.4 Networking

- Tight SRS multiplexing: It is recommended that this function be enabled on all gNodeBs that provide contiguous coverage. Enabling this function only on some gNodeBs in an area is not recommended.
- Other functions: no requirements

3.3.5 Cells

The **NRDUCell.FrequencyBand** parameter must be set to **N38**, **N40**, **N41**, **N77**, **N78**, or **N79**.

Table 3-1 lists the cell requirements of each iBeam function.

Table 3-1 Cell requirements of each iBeam function

Function Name	Supported LampSite Cell Type	Supported Low-Frequency TDD Macro Cell Type	Additional Requirement
Blind interference suppression for SRS measurement	Distributed massive MIMO cells ^a	32T/64T cells	The NRDUCell.NrDuCellNetworkingMode parameter is set to NORMAL_CELL or FUSION_CELL .
Tight SRS multiplexing	Distributed massive MIMO cells ^a	32T/64T cells	None
Downlink multi-BWP interference avoidance	None	32T/64T cells	The NRDUCell.DlBandwidth parameter is set to CELL_BW_60M , CELL_BW_70M , CELL_BW_80M , CELL_BW_90M , or CELL_BW_100M .
Merged scheduling for downlink RLC status reports	Distributed massive MIMO cells ^a	32T/64T cells	None
PDCCH aggregation level reduction	None	32T/64T cells	None
Beam selection optimization	None	32T/64T cells	None

Function Name	Supported LampSite Cell Type	Supported Low-Frequency TDD Macro Cell Type	Additional Requirement
Downlink MU precise scheduling	Distributed massive MIMO cells ^a	32T/64T cells	None
Downlink MU anti-interference scheduling	Distributed massive MIMO cells ^a	32T/64T cells	None
Resource-based adaptive downlink scheduling	Distributed massive MIMO cells ^a	32T/64T cells	None

a: Distributed massive MIMO LampSite cells are those with the **NRDUCell.LampSiteCellFlag** parameter set to **YES** and the **DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter selected.

3.3.6 Others

None

3.4 Operation and Maintenance

3.4.1 Data Configuration

3.4.1.1 Data Preparation

[Table 3-2](#) describes the parameters used for activating iBeam.

Table 3-2 Parameters used for activating iBeam

Parameter Name	Parameter ID	Option	Setting Notes
High Precision Beam Switch	NRDUCellFeatureSw. <i>HighPrecisionBeams witch</i>	None	Set this parameter to ON if iBeam is required.

Table 3-3 describes the parameters used for activating interference suppression enhancement.

Table 3-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
SRS Measurement Optimization Switch	NRDUCellSrsMeas. <i>SrsMeasOptSwitch</i>	SRS_BLIND_IS_MEAS_SW	Select this option if blind interference suppression for SRS measurement is required.
SRS Algorithm Switch	NRDUCellSrs. <i>SrsAlgoSwitch</i>	SRS_TIGHT_MULTIPLEXING_SW	Select this option if tight SRS multiplexing is required.
Downlink Schedule Algorithm Switch	NRDUCellDISch. <i>DISchAlgoSwitch</i>	DL_BWP_HYBRID_INTERFERENCE_AVOIDANCE_SW	Select this option if downlink multi-BWP interference avoidance is required.

Table 3-4 describes the parameters used for activating scheduling enhancement.

Table 3-4 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink Schedule Algorithm Switch	NRDUCellDISch. <i>DISchAlgoSwitch</i>	DL_RLC_STATUS_REPORT_MERGED_SCHEDULING_SW	Select this option if merged scheduling for downlink RLC status reports is required.
PDCCH Algorithm Switch	NRDUCellPdcch. <i>PdcchAlgoSwitch</i>	PDCCH_AGGREGATION_LEVEL_REDUCTION_SW	Select this option if PDCCH aggregation level reduction is required.

Parameter Name	Parameter ID	Option	Setting Notes
Aggregation Level Compression CCE Usage Thld	NRDUCellPdcchAlgo.AggLvlComprCceUsageThld	None	Retain the default value. A larger value of this parameter results in a larger increase in the user-perceived rates of cell-edge UEs but a possible decrease in the average user-perceived rate in the cell. A smaller value of this parameter results in a smaller increase in the user-perceived rates of cell-edge UEs but a possible increase in the average user-perceived rate in the cell.
Channel Optimization Algorithm Switch	NRDUCellBeamAlgo.ChannelOptAlgoSwitch	BEAM_SELECTION_OPT_SWITCH	Select this option if beam selection optimization is required.
DL MU-MIMO Scheduling Supplement Switch	NRDUCellDIMimo.DLMuMimoSchSupplementSw	DL_MU_PRECISE_SCHED_SWITCH	Select this option if downlink MU precise scheduling is required.
DL MU-MIMO Scheduling Supplement Switch	NRDUCellDIMimo.DLMuMimoSchSupplementSw	DL_MU_ANTI_INTERFERENCE_SCHED_SWITCH	Select this option if downlink MU anti-interference scheduling is required.
Downlink Schedule Algorithm Switch	NRDUCellDISch.DISCAlgoSwitch	TAIL_PKT_MCS_OPT_SWITCH	You are advised to select this option for greater benefits if downlink MU precise scheduling or downlink MU anti-interference scheduling is required.
Downlink Schedule Algorithm Switch	NRDUCellDISch.DISCAlgoSwitch	RES_BASED_DL_ADAPT_SCHED_SWITCH	Select this option if resource-based adaptive downlink scheduling is required.

3.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

To enable iBeam:

```
//Enabling iBeam  
MOD NRDUCELLFEATURESW: NrDuCellId=0, HighPrecisionBeamSwitch=ON;
```

To enable functions related to interference suppression enhancement:

```
//Enabling blind interference suppression for SRS measurement  
MOD NRDUCELLSRSMEAS: NrDuCellId=0, SrsMeasOptSwitch=SRS_BLIND_IS_MEAS_SW-1;  
  
//Enabling tight SRS multiplexing  
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=SRS_TIGHT_MULTIPLEXING_SW-1;  
  
//Enabling downlink multi-BWP interference avoidance  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=DL_BWP_HYBRID_INTRF_RANDOM_SW-1;
```

To enable functions related to scheduling enhancement:

```
//Enabling merged scheduling for downlink RLC status reports  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=DL_RLC_STAT_RPT_MERGE_SCH_SW-1;  
  
//Enabling PDCCH aggregation level reduction  
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_AGG_LVL_COMPR_SW-1;  
MOD NRDUCELLPDCCHALGO: NrDuCellId=0, AggLvlComprCceUsageThld=60;  
  
//Enabling beam selection optimization  
MOD NRDUCELLBEAMALGO: NrDuCellId=0, ChannelOptAlgoSwitch=BEAM_SELECT_OPT_SW-1;  
  
//Enabling downlink MU precise scheduling  
MOD NRDUCELLDLMIMO: NrDuCellId=0, DLMuMimoSchSupplementSw=DL_MU_PRECISE_SCH_SW-1;  
  
//Enabling downlink MU anti-interference scheduling  
MOD NRDUCELLDLMIMO: NrDuCellId=0, DLMuMimoSchSupplementSw=DL_MU_ANTI_INTRF_SCH_SW-1;  
  
//(Optional, recommended) Enabling the MCS optimization for tail packets function for downlink MU  
precise scheduling or downlink MU anti-interference scheduling  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=TAIL_PKT_MCS_OPT_SW-1;  
  
//Enabling resource-based adaptive downlink scheduling  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=RES_BASED_DL_ADAPT_SCH_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

To disable functions related to interference suppression enhancement:

```
//Disabling blind interference suppression for SRS measurement  
MOD NRDUCELLSRSEAS: NrDuCellId=0, SrsMeasOptSwitch=SRS_BLIND_IS_MEAS_SW-0;  
  
//Disabling tight SRS multiplexing  
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=SRS_TIGHT_MULTIPLEXING_SW-0;  
  
//Disabling downlink multi-BWP interference avoidance  
MOD NRDUCELLDLSCHE: NrDuCellId=0, DISchAlgoSwitch=DL_BWP_HYBRID_INTRF_RANDOM_SW-0;
```

To disable functions related to scheduling enhancement:

```
//Disabling merged scheduling for downlink RLC status reports  
MOD NRDUCELLDLSCHE: NrDuCellId=0, DISchAlgoSwitch=DL_RLC_STAT_RPT_MERGE_SCH_SW-0;  
  
//Disabling PDCCH aggregation level reduction  
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_AGG_LVL_COMPR_SW-0;  
  
//Disabling beam selection optimization  
MOD NRDUCELLBEAMALGO: NrDuCellId=0, ChannelOptAlgoSwitch=BEAM_SELECT_OPT_SW-0;  
  
//(Optional) Disabling MCS optimization for tail packets  
MOD NRDUCELLDLSCHE: NrDuCellId=0, DISchAlgoSwitch=TAIL_PKT_MCS_OPT_SW-0;  
  
//Disabling downlink MU precise scheduling  
MOD NRDUCELLDLMIMO: NrDuCellId=0, DLMuMimoSchSupplementSw=DL_MU_PRECISE_SCH_SW-0;  
  
//Disabling downlink MU anti-interference scheduling  
MOD NRDUCELLDLMIMO: NrDuCellId=0, DLMuMimoSchSupplementSw=DL_MU_ANTI_INTRF_SCH_SW-0;  
  
//Disabling resource-based adaptive downlink scheduling  
MOD NRDUCELLDLSCHE: NrDuCellId=0, DISchAlgoSwitch=RES_BASED_DL_ADAPT_SCH_SW-0;
```

To disable iBeam:

```
//Disabling iBeam  
MOD NRDUCELLFEATURESW: NrDuCellId=0, HighPrecisionBeamSwitch=OFF;
```

3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

3.4.2 Activation Verification

Observe the **User Downlink Average Throughput (DU)** before and after iBeam is enabled. If the value of this indicator increases, iBeam has taken effect.

3.4.3 Network Monitoring

After iBeam is enabled, use the performance indicators described in [3.2 Network Analysis](#) to monitor cell performance.

4 iBeam 2.0

4.1 Principles

iBeam 2.0 has been introduced to further increase the downlink user-perceived rate in cells that offer contiguous coverage. This feature applies to TDD massive MIMO cells. It improves downlink performance through weight enhancement, scheduling enhancement, and interference-aware scheduling enhancement.

To enable iBeam 2.0 and its various functions for these enhancements, the **NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw** parameter must be set to ON.

4.1.1 Weight Enhancement

PDSCH Robust Weights

Low weight accuracy is common during initial access and handovers because these procedures typically use precoding matrix indication (PMI)-based weights. **To enable more accurate beamforming toward UEs during these procedures, PDSCH robust weights can be used. These weights are generated using SRS-based weights to increase the downlink user-perceived rate.**

The PDSCH robust weights function is controlled by the **DL_ROBUST_WEIGHT_SW** option of the **NRDUCellPdschPrecode.PrecodingAlgoSwitch** parameter.

Adaptive Filtering for SRS Channel Estimation

The accuracy and validity of SRS measurement determine downlink performance. Adaptive filtering for SRS channel estimation adaptively reduces the noise for SRS in massive MIMO cells. The noise reduction mitigates SRS interference and improves SRS quality and weight accuracy, thereby increasing the downlink user-perceived rate.

Adaptive filtering for SRS channel estimation is controlled by the **CHN_EST_ADAPT_FLTR_SW** option of the **NRDUCellSrs.SrsDetectionAlgoSwitch** parameter. This function depends on a time-domain channel correlation threshold

(specified by **NRDUCellSrsMeas.TimeChnCorrThld**) to take effect. A lower threshold results in a higher probability that this function takes effect in a cell, and a higher threshold results in a lower probability.

4.1.2 Scheduling Enhancement

Rank Optimization for Cell-Edge UEs

Both ranks and MCSs determine the spectral efficiency in downlink scheduling for MU-MIMO. To maximize spectral efficiency, rank optimization for cell-edge UEs selects suitable ranks and MCSs for cell-edge MU-MIMO UEs that would otherwise use low-index MCSs. This increases the downlink user-perceived rate.

Rank optimization for cell-edge UEs is controlled by the **FAR_UE_RANK_OPT_SW** option of the **NRDUCellDlMimo.DLMuMimoSchSupplementSw** parameter.

Precise MU-MIMO Evaluation

Transmit power and selected MCSs may be inaccurate when RBs with different lengths are paired. Precise MU-MIMO evaluation adjusts power allocation and MCS selection for MU-MIMO UEs to increase the proportion of MU-MIMO UEs and thereby increase the downlink user-perceived rate.

Precise MU-MIMO evaluation is controlled by the **PRECISE_MUMIMO_EVAL_SW** option of the **NRDUCellDlMimo.DLMuMimoSchSupplementSw** parameter. When this function is enabled, it is recommended that MCS optimization for tail packets also be enabled for greater benefits. (MCS optimization for tail packets is controlled by the **TAIL_PKT_MCS_OPT_SW** option of the **NRDUCellDlSch.DlSchAlgoSwitch** parameter.)

Target IBLER Adjustment in MU-MIMO

MU-MIMO features rapid and dynamic changes in pairing relationships. This means that the target IBLER must be adjusted independently for MU-MIMO to select MCSs more appropriately. This function adjusts the target IBLER in MU-MIMO to increase the downlink user-perceived rate of MU-MIMO UEs.

This "target IBLER adjustment in MU-MIMO" function is controlled by the **MUMIMO_IBLER_TARGET_ADJ_SW** option of the **NRDUCellDlMimo.DLMuMimoSchSupplementSw** parameter. When this option is selected, the target IBLER in downlink MU-MIMO can be adjusted based on the value of the **NRDUCellDlMimo.MuMimoIblerTarget** parameter. When this function is enabled, it is recommended that MCS optimization for tail packets also be enabled for greater benefits. (MCS optimization for tail packets is controlled by the **TAIL_PKT_MCS_OPT_SW** option of the **NRDUCellDlSch.DlSchAlgoSwitch** parameter.)

Dynamic Clustering-based Grouping

Dynamic clustering-based grouping dynamically groups UEs based on the clustering algorithm and correlations between UEs. This function allows a larger proportion of UEs to be paired in the downlink in a cell and thereby increases the downlink user-perceived rate.

Dynamic clustering-based grouping is enabled when the **NRDUCellPdsch.DLMuMimoGroupMode** parameter is set to **DYNAMIC_CLUSTER_GROUP**. When this function is enabled, it is recommended that MCS optimization for tail packets also be enabled for greater benefits. (MCS optimization for tail packets is controlled by the **TAIL_PKT_MCS_OPT_SW** option of the **NRDUCellDISch.DISchAlgoSwitch** parameter.)

Downlink Correlation Acceleration

The base station needs to obtain correlations between UEs in a cell for MU pairing. Downlink correlation acceleration improves the base station's capability to calculate correlations between UEs. With more correlations obtained, the MU-MIMO pairing rate and the cell throughput will increase.

Downlink correlation acceleration is controlled by the **DL_CORR_ACCELERATION_SW** option of the **NRDUCellDlMimo.DLMuMimoSchSupplementSw** parameter.

The **NRDUCellDlMimo.DlMuCorrAcceleratorMode** parameter specifies the accelerator mode for this function:

- If this parameter is set to **LIVE**, live calculation mode is used.
- If this parameter is set to **PRECALCULATION**, precalculation mode is used. In this case, correlations between to-be-paired UEs are calculated in advance, increasing the correlation-based pairing rate.

CCE Partial Symbol Adaptation

CCE partial symbol adaptation dynamically adjusts the number of CCE symbols based on the downlink slot type. The adjustment increases the spectral efficiency in downlink scheduling and the average downlink UE throughput in a cell. This function is controlled by the **CCE_PARTIAL_SYM_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter.

When this function is enabled, you are advised to enable PDCCH blind detection assignment optimization (controlled by the **PDCCH_BLIND_DET_ASSIGN_OPT_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter) as well. The purpose is to balance blind detection times across different PDCCH aggregation levels, so as to increase uplink and downlink CCE allocation success rates.

Certain UEs are incompatible with CCE partial symbol adaptation. To address this compatibility issue, it is recommended that blacklist control for dynamic PDCCH symbol allocation be enabled by selecting the **PDCCH_SYM_DYNAMIC_ALLOC_SW_OFF** option of the **gNBUECompatSw.BlacklistCtrlExtSwitch** parameter. This blacklist control function disables CCE partial symbol adaptation for such UEs.

Downlink Scheduling Based on Load and Experience

In a heavily loaded cell, data packet models vary greatly between UEs and the transmission quality fluctuates significantly over the air interface. This will result in highly unstable downlink transmission delay of UEs and compromise the cell's overall transmission delay. To address this issue, downlink scheduling based on load and experience has been introduced. This function determines the scheduling

priorities of UEs based on the data packet models and transmission quality over the air interface. The aim is to minimize the transmission delay of UEs in a cell and increase the downlink user-perceived rate. This function is controlled by the **LOAD_BASED_DL_EXP_SCH_SW** option of the **NRDUCellDISch.DISchAlgoSwitch** parameter.

The **NRDUCellDISch.DIAdaptSchPrbThld** parameter specifies the PRB usage threshold for this function. This function takes effect only if there are a low proportion of slots where the PRB usage exceeds this threshold in a measurement period.

4.1.3 Interference-aware Scheduling Enhancement

Inter-Cell Interference Avoidance

In intra-frequency networking, downlink user experience is affected if UEs scheduled simultaneously in different cells interfere with each other. To address this issue, inter-cell interference avoidance has been introduced. This function preferentially selects UEs that cause low interference to each other for simultaneous scheduling to reduce inter-cell interference. Inter-cell interference avoidance is controlled by the **INTER_CELL_INTRF_AVOID_SW** option of the **NRDUCellCollabServ.MultiCellMimoSwitch** parameter. The **NRDUCellCollabServ.IntrfAvoidDlTargetIbler** parameter specifies the target IBLER for UEs involved in this function.

Precise Neighboring Cell Measurement for SRS IC

SRS interference changes rapidly and constantly in cells providing contiguous coverage. In response, precise neighboring cell measurement for SRS interference cancellation (IC) enables the base station to measure the SRS interference from neighboring cells in real time and select the neighboring cell causing the strongest SRS interference for IC. In this way, SRS measurement is more accurate.

Precise neighboring cell measurement for SRS IC is controlled by the **SRS_IC_NCELL_PRECISE_MEAS_SW** option of the **NRDUCellSrs.SrsDetectionAlgoSwitch** parameter.

CSI-RS Interference Coordination

Channel state information reference signal (CSI-RS) in cells providing contiguous coverage may be configured to use the same time-frequency resources. This increases the probability of CSI-RS interference between neighboring cells. Such interference results in inaccurate downlink channel state information (CQI, RI, and PMI) reported by UEs and reduces downlink transmission efficiency. To mitigate these issues, CSI-RS interference coordination has been introduced. This function staggers time-frequency resources used by CSI-RS between cells in the time domain to reduce CSI-RS interference from neighboring cells.

CSI-RS interference coordination is controlled by the **CSIRS_INTRF_COORD_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter.

4.2 Network Analysis

4.2.1 Benefits

iBeam 2.0 increases the **User Downlink Average Throughput (DU)** and downlink average MCS index (equal to $\text{SUM}(k \times \text{N.ChMeas.PDSCH.MCS}.k) / \text{SUM}(\text{N.ChMeas.PDSCH.MCS}.k)$, where k ranges from 0 to 28). The actual benefits of iBeam 2.0 depend on factors such as the traffic model, network load, and PCI planning. The higher the downlink PRB usage and the stronger the SRS interference from neighboring cells, the greater the benefits. iBeam 2.0 is recommended in intra-frequency contiguous massive MIMO networking where all the following conditions are met:

- If there are no fixed wireless access (FWA) users, the conditions are as follows:
 - The downlink PRB usage (equal to $\text{N.PRB.DL.Used.Avg} / \text{N.PRB.DL.Avail.Avg}$) ranges from 30% to 70%.
 - The SRS interference (indicated by the **N.SRS.NI.Avg** counter) from neighboring cells is greater than -105 dBm.
 - The proportion of PRBs paired for downlink MU-MIMO is greater than 10%. This proportion equals $\text{N.ChMeas.MIMO.DL.Pair.PR} / (\text{N.PRB.DL.Used.Avg} \times \text{N.DL.PDSCH.Tti.Num}) \times 100\%$.
 - The proportion of SRS SINRs before equalization (SRS PreSINRs) less than 0 dB is greater than 50%. This proportion equals $(\text{N.UL.SRS.PreSINR.Index0} + \text{N.UL.SRS.PreSINR.Index1} + \text{N.UL.SRS.PreSINR.Index2} + \text{N.UL.SRS.PreSINR.Index3}) / (\text{N.UL.SRS.PreSINR.Index0} + \text{N.UL.SRS.PreSINR.Index1} + \text{N.UL.SRS.PreSINR.Index2} + \text{N.UL.SRS.PreSINR.Index3} + \text{N.UL.SRS.PreSINR.Index4} + \text{N.UL.SRS.PreSINR.Index5})$.
 - The maximum number of downlink paired layers in a cell is greater than or equal to 8.
- If there are FWA users, the conditions are as follows:
 - The downlink PRB usage (equal to $\text{N.PRB.DL.Used.Avg} / \text{N.PRB.DL.Avail.Avg}$) ranges from 50% to 70%.
 - The SRS interference (indicated by the **N.SRS.NI.Avg** counter) from neighboring cells is greater than -105 dBm.
 - The proportion of PRBs paired for downlink MU-MIMO is greater than 15%. This proportion equals $\text{N.ChMeas.MIMO.DL.Pair.PR} / (\text{N.PRB.DL.Used.Avg} \times \text{N.DL.PDSCH.Tti.Num}) \times 100\%$.
 - The proportion of SRS PreSINRs less than 0 dB is greater than 50%. This proportion equals $(\text{N.UL.SRS.PreSINR.Index0} + \text{N.UL.SRS.PreSINR.Index1} + \text{N.UL.SRS.PreSINR.Index2} + \text{N.UL.SRS.PreSINR.Index3}) / (\text{N.UL.SRS.PreSINR.Index0} + \text{N.UL.SRS.PreSINR.Index1} + \text{N.UL.SRS.PreSINR.Index2} + \text{N.UL.SRS.PreSINR.Index3} + \text{N.UL.SRS.PreSINR.Index4} + \text{N.UL.SRS.PreSINR.Index5})$.
 - The maximum number of downlink paired layers in a cell is greater than or equal to 8.

If the preceding conditions are not met, some functions offer fewer or even no benefits. In addition, iBeam 2.0 offer fewer benefits if any of the functions for weight enhancement, scheduling enhancement, and interference-aware scheduling enhancement is disabled.

For greater benefits of precise MU-MIMO evaluation, target IBLER adjustment in MU-MIMO, or dynamic clustering-based grouping, it is recommended that MCS optimization for tail packets also be enabled. (MCS optimization for tail packets is controlled by the **TAIL_PKT_MCS_OPT_SW** option of the **NRDUCellDLSch.DLSchAlgoSwitch** parameter.)

4.2.2 Impacts

The impacts between this function and other functions are described in [4.2.2 Impacts](#).

iBeam 2.0 may have the following impacts on the network:

- The downlink MCS index may increase.
- The MU-MIMO pairing rate may increase. For details about the impacts of the increase on related KPIs, see the "Impacts" section of MU-MIMO (low-frequency TDD) in *MIMO (TDD)*. (MU-MIMO pairing rate = $\frac{\text{N.ChMeas.MIMO.DL.Pair.PRB}}{(\text{N.PR.BD.LUsed.Avg} \times \text{N.DL.PDSCH.Tti.Num})}$)
- The first packet delay (indicated by the **N.Traffic.DL.RlcFirstPktDelay.Time** counter) may increase.
- The service drop rate in a cell may change. (Service drop rate in a cell = $\frac{\text{N.UECntx.AbnormRel}}{(\text{N.UECntx.NormRel} + \text{N.UECntx.AbnormRel})}$)
- The downlink PRB usage may decrease. (Downlink PRB usage = $\frac{\text{N.PR.BD.LUsed.Avg}}{\text{N.PR.BD.LAvail.Avg}}$)
- The proportion of samples in which the average downlink UE throughput in a cell falls within low-throughput ranges may change. (This proportion = $\frac{(\text{N.Thp.DL.Samp.Index0} + \text{N.Thp.DL.Samp.Index1})}{\text{SUM}(\text{N.Thp.DL.Samp.Index}k)}$, where k is in the range of 0 to 14)
- The average number of layers paired for downlink MIMO (indicated by the **N.ChMeas.MIMO.DL.Pair.Layer.Avg** counter) may change.
- The average SRS interference (indicated by the **N.SRS.NI.Avg** counter) may change.
- The values of the following scheduling-related indicators may change:
 - Downlink average rank, which equals $(1 \times \text{N.PDSCH.InitTbDL.Rank1} + 2 \times \text{N.PDSCH.InitTbDL.Rank2} + 3 \times \text{N.PDSCH.InitTbDL.Rank3} + 4 \times \text{N.PDSCH.InitTbDL.Rank4}) / (\text{N.PDSCH.InitTbDL.Rank1} + \text{N.PDSCH.InitTbDL.Rank2} + \text{N.PDSCH.InitTbDL.Rank3} + \text{N.PDSCH.InitTbDL.Rank4})$
 - Downlink IBLER, which equals $(\text{N.DL.SCH.QPSK.ErrTB.Ibler} + \text{N.DL.SCH.16QAM.ErrTB.Ibler} + \text{N.DL.SCH.64QAM.ErrTB.Ibler} + \text{N.DL.SCH.256QAM.ErrTB.Ibler}) / (\text{N.DL.SCH.QPSK.TB} + \text{N.DL.SCH.16QAM.TB} + \text{N.DL.SCH.64QAM.TB} + \text{N.DL.SCH.256QAM.TB})$
 - Downlink RBLE, which equals $(\text{N.DL.SCH.QPSK.ErrTB.Rbler} + \text{N.DL.SCH.16QAM.ErrTB.Rbler} + \text{N.DL.SCH.64QAM.ErrTB.Rbler} + \text{N.DL.SCH.256QAM.ErrTB.Rbler}) / (\text{N.DL.SCH.QPSK.TB} + \text{N.DL.SCH.16QAM.TB} + \text{N.DL.SCH.64QAM.TB} + \text{N.DL.SCH.256QAM.TB})$
 - Downlink padding rate, which equals $1 - (\text{N.ThpVol.DL} / \text{N.ThpVol.DL.Cell})$

- The downlink PDCCH CCE usage, equivalent PDCCH CCE usage, and average uplink and downlink CCE aggregation levels of a cell may change.
 - Downlink PDCCH CCE usage of a cell = $\text{N.CCE.DLDCI.Used.Equivalent} / \text{N.CCE.Avail.Equivalent}$
 - Equivalent PDCCH CCE usage of a cell = $(\text{N.CCE.ULDCI.Used.Equivalent} + \text{N.CCE.DLDCI.Used.Equivalent}) / \text{N.CCE.Avail.Equivalent}$
 - Average uplink CCE aggregation level = $(1 \times \text{N.CCE.UL.AggLvl1Num} + 2 \times \text{N.CCE.UL.AggLvl2Num} + 4 \times \text{N.CCE.UL.AggLvl4Num} + 8 \times \text{N.CCE.UL.AggLvl8Num} + 16 \times \text{N.CCE.UL.AggLvl16Num}) / (\text{N.CCE.UL.AggLvl1Num} + \text{N.CCE.UL.AggLvl2Num} + \text{N.CCE.UL.AggLvl4Num} + \text{N.CCE.UL.AggLvl8Num} + \text{N.CCE.UL.AggLvl16Num})$
 - Average downlink CCE aggregation level = $(1 \times \text{N.CCE.DL.AggLvl1Num} + 2 \times \text{N.CCE.DL.AggLvl2Num} + 4 \times \text{N.CCE.DL.AggLvl4Num} + 8 \times \text{N.CCE.DL.AggLvl8Num} + 16 \times \text{N.CCE.DL.AggLvl16Num}) / (\text{N.CCE.DL.AggLvl1Num} + \text{N.CCE.DL.AggLvl2Num} + \text{N.CCE.DL.AggLvl4Num} + \text{N.CCE.DL.AggLvl8Num} + \text{N.CCE.DL.AggLvl16Num})$

Downlink scheduling based on load and experience may have the following impacts:

- The DU delay may increase. (DU delay (ms) = $(\text{N.QoS.DL.PktDelayAirInterface.Time} / \text{N.QoS.DL.PktDelayAirInterface.Num} + \text{N.QoS.DL.PktDelaygNBdu.Time} / \text{N.QoS.DL.PktDelaygNBdu.Num}) / 1000$)
- The average downlink cell throughput may change.
- The number of active UEs in the uplink (indicated by the **N.User.RRConn.Active.UL.Avg** counter) may change.
- The number of PRBs paired for downlink MIMO (indicated by the **N.ChMeas.MIMO.DL.Pair.PRB** counter) may change.
- The number of PDCCH CCEs may change. The following indicators are involved:
 - Average number of used PDCCH CCEs (indicated by the **N.CCE.Used.Avg** counter)
 - Average number of PDCCH CCEs used for downlink DCI (indicated by the **N.CCE.DLDCI.Used.Avg** counter)
 - Average number of PDCCH CCEs used for uplink DCI (indicated by the **N.CCE.ULDCI.Used.Avg** counter)
- The average uplink received signal strength indicator (RSSI) of channels (indicated by the **N.UL.RSSI.Avg** counter) may change.

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	NE	Sales Unit
FOFD-091200	iBeam 2.0	NR0S0DLEH R00	gNodeB	Per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

- PDSCH robust weights

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 2.0	NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	PDSCH robust weights can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw parameter is set to ON .

- Adaptive filtering for SRS channel estimation

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 2.0	NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Adaptive filtering for SRS channel estimation can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw parameter is set to ON .
Low-frequency TDD	Low-speed cell	NRDUCell.HighSpeedFlag set to LOW_SPEED	None	Adaptive filtering for SRS channel estimation can be enabled only for a low-speed cell.
Low-frequency TDD	Adaptive filtering for SRS channel estimation	CHN_EST_ADAPT_FLTR_SW option of the NRDUCellSrs.SrsDetectionAlgorithmSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	The NRDUCellSrsMeas.TimeChnCorrectionThld parameter can take effect only when the CHN_EST_ADAPT_FLTR_SW option of the NRDUCellSrs.SrsDetectionAlgorithmSwitch parameter is selected.

- Rank optimization for cell-edge UEs

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 2.0	NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Rank optimization for cell-edge UEs can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw parameter is set to ON .
Low-frequency TDD	Spectral-efficiency-based adaptive switching between SU-MIMO and MU-MIMO	DL_SE_SU_MU_MIMO_ADAPT_SW option of the NRDUCellDimmo.DLMuMimoSchSupplementSw parameter	<i>MIMO (TDD)</i>	Rank optimization for cell-edge UEs can take effect only when spectral-efficiency-based adaptive switching between SU-MIMO and MU-MIMO is enabled.

- Precise MU-MIMO evaluation

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 2.0	NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Precise MU-MIMO evaluation can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw parameter is set to ON .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Spectral-efficiency-based adaptive switching between SU-MIMO and MU-MIMO	DL_SE_SU_MU_MIMO_ADAPT_SW option of the NRDUCellDimmo.DLMuMimoSchSupplementSw parameter	<i>MIMO (TDD)</i>	Precise MU-MIMO evaluation can take effect only when spectral-efficiency-based adaptive switching between SU-MIMO and MU-MIMO is enabled.

- Target IBLER adjustment in MU-MIMO

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 2.0	NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Target IBLER adjustment in MU-MIMO can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw parameter is set to ON .
Low-frequency TDD	Downlink MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Target IBLER adjustment in MU-MIMO can take effect only when downlink MU-MIMO is enabled.

- Dynamic clustering-based grouping

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 2.0	NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Dynamic clustering-based grouping can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw parameter is set to ON .

- Downlink correlation acceleration

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 2.0	NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Downlink correlation acceleration can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw parameter is set to ON .
Low-frequency TDD	Downlink MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Downlink correlation acceleration can take effect only when downlink MU-MIMO is enabled.

- CCE partial symbol adaptation

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 2.0	NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	CCE partial symbol adaptation can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw parameter is set to ON .
Low-frequency TDD	PDCCH symbol number adaptation	UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter	<i>Channel Management</i>	CCE partial symbol adaptation can take effect only when the UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter is selected.
Low-frequency TDD	Maximum Aggregated PDCCH Power Offset	NRDUCellChnPower.MaxPdcchAggPwrOffset	None	CCE partial symbol adaptation can take effect only when the NRDUCellChnPower.MaxPdcchAggPwrOffset parameter is set to a value greater than or equal to 30.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink Bandwidth	NRDUCell.DlBandwidth	None	CCE partial symbol adaptation can take effect only when each of the following conditions is met: - The NRDUCell.DlBandwidth parameter is set to CELL_BW_40M, CELL_BW_50M, CELL_BW_60M, CELL_BW_70M, CELL_BW_80M, CELL_BW_90M, or CELL_BW_100M. - The LTE_NRTD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter is selected. $106 \leq (\text{gNBDULteNrSptShrCg.NrReservedRbEndIndex} - \text{gNBDULteNrSptShrCg.NrReservedRbStartIndex})$
Low-frequency TDD	LTE TDD and NR Spectrum Sharing Switch	LTE_NRTD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	None	
Low-frequency TDD	NR Reserved RB Start Index	gNBDULteNrSptShrCg.NrReservedRbStartIndex	None	
Low-frequency TDD	NR Reserved RB End Index	gNBDULteNrSptShrCg.NrReservedRbEndIndex	None	
Low-frequency TDD	Common Control Resource RB Number	NRDUCellCoreSet.CommonCtrlResRbNum	None	

RAT	Function Name	Function Switch	Reference	Description
				<i>RbStartIndex + 1</i> ≤ 189 The NRDUCellCoreset.CommonCtrlResRbNum parameter is set to RB48.
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode NRDUCellMultiTrp.DataTransMode	<i>Cell Combination</i>	CCE partial symbol adaptation can take effect only when each of the following conditions is met:
Low-frequency TDD	Common Control Resource RB Number	NRDUCellCoreset.CommonCtrlResRbNum	None	- The NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL_COMBINE_MODE. - The NRDUCellMMultiTrp.DataTransMode parameter is set to BASIC_MODE or DPS_MODE. - The NRDUCellCoreset.CommonCtrlResRbNum parameter is set to RB48.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Smart PDCCH symbol allocation	PDCCH_SYM_SMART_ALLOC_SW option of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter	None	PDCCH blind detection assignment optimization can take effect only when any of the following options is selected: the PDCCH_SYM_SMART_ALLOC_SW , MULTI_SEARCH_SPACE_SW , and CCE_RESOURCE_OPT_SW options of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter, and the CCE_PARTIAL_SYM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter.
Low-frequency TDD	Configuration of multiple PDCCH search spaces	MULTI_SEARCH_SPACE_SW option of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter	<i>Channel Management</i>	
Low-frequency TDD	CCE resource optimization	CCE_RESOURCE_OPT_SW option of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter	<i>MIMO (TDD)</i>	
Low-frequency TDD	CCE partial symbol adaptation	CCE_PARTIAL_SYM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	

- Downlink scheduling based on load and experience

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 2.0	NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Downlink scheduling based on load and experience can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw parameter is set to ON .
Low-frequency TDD	Downlink scheduling based on load and experience	LOAD_BASED_DL_EXP_SCH_SW option of the NRDUCellDisch.DLSchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	The NRDUCellDisch.DLAdaptSchPrbThld parameter can take effect only when downlink scheduling based on load and experience is enabled.
Low-frequency TDD	Resource-based adaptive scheduling for massive MIMO	RES_BASED_MM_ADAPT_SCH_SW option of the NRDUCellServiceExp.ServiceExpAlgoSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	Downlink scheduling based on load and experience can take effect only when resource-based adaptive scheduling for massive MIMO is in effect.

- Inter-cell interference avoidance

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 2.0	NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Inter-cell interference avoidance can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw parameter is set to ON .
Low-frequency TDD	UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>Channel Management</i>	UE-characteristic-based SRS period adaptation must be enabled before inter-cell interference avoidance is enabled.
Low-frequency TDD	Downlink precise RB estimation	PRECISE_DL_RB_ESTIMATE_SW option of the NRDUCellDLSch.DLSchAlgoSwitch parameter	<i>Downlink Scheduling</i>	Inter-cell interference avoidance can take effect only when downlink precise RB estimation is enabled.

- Precise neighboring cell measurement for SRS IC

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 2.0	NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Precise neighboring cell measurement for SRS IC can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw parameter is set to ON .
Low-frequency TDD	SRS IC	SRS_IC_SW option of the NRDUCellSrs.SrsDetectionAlgorithmSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	SRS IC must be enabled before precise neighboring cell measurement for SRS IC is enabled.

- CSI-RS interference coordination

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 2.0	NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	CSI-RS interference coordination can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw parameter is set to ON .
Low-frequency TDD	CSIRS Cell Resource Number	NRDUCellCsirs.CsirsCellResourceNum	None	When CSI-RS interference coordination is enabled, CSIRS Cell Resource Number must be set to 2_RESOURCE or 4_RESOURCE .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CSI Period	NRDUCellCsirs.CsiPeriod	None	- When CSI-RS interference coordination is enabled and the NRDUCellCsirs.CsirsCellResourceNum parameter is set to 4_RESOURCE, CSI Period must be set to one of the following values: SLOT160 SLOT320 - When CSI-RS interference coordination is enabled and the NRDUCellCsirs.CsirsCellResourceNum parameter is set to 2_RESOURCE, CSI Period must be set to one of the following values: SLOT80 SLOT160 SLOT320
Low-frequency TDD	NCell CSI-RS Interference Avoidance	NRDUCellCsirs.NCellCsirsIntrfAvoid	None	When CSI-RS interference coordination is enabled, NCell CSI-RS Interference Avoidance must be set to one of the following values: P_CELL P_CELL_AND_AP P_CELL_AND_P_USER P_CELL_AND_P_USER_AND_AP

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CSI Period	NRDUCellCsirs.CsiPeriod	None	CSI Period must be set to SLOT160 or SLOT320 when CSI-RS interference coordination is enabled in any of the following scenarios: • Scenario 1: Each of the following conditions is met: – Among the options of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter, only the UL_CELL_LEVEL_INTERF_AVOID and DL_UE_LEVEL_INTERF_AVOID options are selected. – The NRDUCellIntrfIdent.IntrfIdentificationMethod parameter is set to UL_DL_MANUAL. – At least one NRDUCellIntrfBand MO has the NRDUCellIntrfBand.InterferenceDirection parameter set to UPLINK_DOWNLINK. – The NRDUCellCsirs.CsirsCellResourceNum parameter is set to 2_RESOURCE. • Scenario 2: Each of the following conditions is met: – The BI_CARRIER_SW

RAT	Function Name	Function Switch	Reference	Description
				option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter or the PMI_WEIGHT_OPT_SW option of the NRDUCellBeamAlgo.WeightAlgoSwitch parameter is selected. - The NRDUCellCsirs.CsirsCellResourceNum parameter is set to 2_RESOURCE. CSI Period must be set to SLOT320 when CSI-RS interference coordination is enabled in any of the following scenarios: • Scenario 1: Each of the following conditions is met: - Among the options of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter, only the UL_CELL_LEVEL_INTERF_AVOID and DL_UE_LEVEL_INTERF_AVOID options are selected. - The NRDUCellIntrfIdent.IntrfIdentificationMethod parameter is set to UL_DL_MANUAL. - At least one NRDUCellIntrfBand MO has the NRDUCellIntrfBand

RAT	Function Name	Function Switch	Reference	Description
				d.InterferenceDirection parameter set to UPLINK_DOWNLINK. - The NRDUCellCsirs.CsirsCellResourceNum parameter is set to 4_RESOURCE. • Scenario 2: Each of the following conditions is met: - The BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter or the PMI_WEIGHT_OPT_SW option of the NRDUCellBeamAlgo.WeightAlgoSwitch parameter is selected. - The NRDUCellCsirs.CsirsCellResourceNum parameter is set to 4_RESOURCE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CSI Period	NRDUCellCsirs.CsiPeriod	None	CSI Period must be set to SLOT80, SLOT160, or SLOT320 when CSI-RS interference coordination is enabled and each of the following conditions is met: - The NRDUCell.AerialCellF_{lag} parameter is set to GROUND_AERIAL_CELL. - The PMI_WEIGHT_OPT_SW option of the NRDUCellBeamAlgo.WeightAlgoSwitch parameter is deselected. - The NRDUCellCsirs.CsirsCellResourceNum parameter is set to 2_RESOURCE. CSI Period must be set to SLOT160 or SLOT320 when CSI-RS interference coordination is enabled and each of the following conditions is met: - The NRDUCell.AerialCellF_{lag} parameter is set to GROUND_AERIAL_CELL. - The PMI_WEIGHT_OPT_SW option of the NRDUCellBeamAlgo.WeightAlgoSwitch parameter is deselected.

RAT	Function Name	Function Switch	Reference	Description
				The NRDUCellCsirs.CsirsCellResourceNum parameter is set to 4_RESOURCE.

4.3.2.2 Mutually Exclusive Functions

- PDSCH robust weights

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SCC SRS-Only Weight Switch	SCC_SRS_ONLY_WEIGHT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter	None	SCC SRS-Only Weight Switch can be turned on only when the PDSCH robust weights function is disabled.
Low-frequency TDD	DL MU Intrf Reject Precoding Policy	NRDUCellPdsc hPrecode.DIM uIRPrecodePol set to MU_ALLUSER_LOWLATENCY or MU_ALLUSER_POWER_ENH	None	With the DL_MU_NLOS_IR_PRECODING_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter selected, PDSCH robust weights can be enabled only when the NRDUCellPdsc hPrecode.DIM uIRPrecodePol parameter is set to MU_ALLUSER_LOWLATENCY or MU_ALLUSER_POWER_ENH .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL High-Speed Robust Weight Sw	DL_HS_ROBUST_WEIGHT_SW option of the NRDUCellHighSpeedMob.HighSpeedAlgoOptSw parameter	None	DL High-Speed Robust Weight Sw must be turned off when the PDSCH robust weights function is enabled.

- Dynamic clustering-based grouping

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink MU-MIMO layer isolation	DL_MU_MIMO_ISOLATION_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	Downlink MU-MIMO layer isolation cannot be enabled together with dynamic clustering-based grouping.

- CCE partial symbol adaptation

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Smart downlink repetition	DL_SMART_REPETITION_SW option of the NRDUCellQciBearer.ServiceDiffAlgoSwitch parameter	None	Smart downlink repetition cannot be enabled together with CCE partial symbol adaptation.
Low-frequency TDD	Bi-Carrier	BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	None	Bi-Carrier cannot be enabled together with CCE partial symbol adaptation.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast CBF	QUICK_CBF_SW option of the NRDUCellComp.CsCbfAlgoSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	Fast CBF cannot be enabled together with CCE partial symbol adaptation.
Low-frequency TDD	Single DCI	DL_SINGLE_DC_I_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	Single downlink control information (DCI) cannot be enabled together with CCE partial symbol adaptation.
Low-frequency TDD	Grid-specific power control in the smart power lock	AVERAGE_3DEIRP_CONTROL_SW option of the NRDUCellSmartPwrLock.SmartPowerLockSw parameter	None	Grid-specific power control in the smart power lock cannot be enabled together with CCE partial symbol adaptation.
Low-frequency TDD	Downlink semi-persistent scheduling	DL_SPS_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	Downlink semi-persistent scheduling cannot be enabled together with CCE partial symbol adaptation.
Low-frequency TDD	Super uplink with independent SUL deployment	SUPER_UL_TDM_SCH_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	Super uplink with independent SUL deployment cannot be enabled together with CCE partial symbol adaptation.
Low-frequency TDD	Super uplink with SUL and NR FDD co-deployment	FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	Super uplink with SUL and NR FDD co-deployment cannot be enabled together with CCE partial symbol adaptation.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-base-station super uplink	INTER_GNB_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	None	Inter-base-station super uplink cannot be enabled together with CCE partial symbol adaptation.
Low-frequency TDD	Downlink Bandwidth	NRDUCell.DlBandwidth	None	CCE partial symbol adaptation can take effect only when each of the following conditions is met:
Low-frequency TDD	LTE TDD and NR dynamic spectrum sharing	LTE_NR_TDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	None	- The NRDUCell.DlBandwidth parameter is set to CELL_BW_40M, CELL_BW_50M, CELL_BW_60M, or CELL_BW_70M. - The LTE_NR_TDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter is deselected. - The NRDUCellCoreset.CommonCtrlResRbNum parameter is set to RB48.
Low-frequency TDD	Common Control Resource RB Number	NRDUCellCoreset.CommonCtrlResRbNum	None	

- Inter-cell interference avoidance

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	Inter-cell interference avoidance cannot be enabled in high-speed scenarios.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	U-TDOA-based high-precision positioning/TOA fingerprint-based positioning	UTDOA_SW option of the NRCellAlgoSwitch.LcsSwitch parameter	None	U-TDOA-based high-precision positioning or TOA fingerprint-based positioning cannot be enabled together with inter-cell interference avoidance.
Low-frequency TDD	SRS field strength-based positioning	SRS_RSRP_LOCATION_SW option of the NRCellAlgoSwitch.LcsSwitch parameter	None	SRS field strength-based positioning cannot be enabled together with inter-cell interference avoidance.
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	Distributed massive MIMO cannot be enabled together with inter-cell interference avoidance.
Low-frequency TDD	VMIMO in DAS-based indoor coverage	NRDUCellAlgoSwitch.VmimoSwitch	<i>Indoor Coverage</i>	VMIMO in DAS-based indoor coverage cannot be enabled together with inter-cell interference avoidance.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	IAB deployment	NRDUCell.Depl oyPosition set to DEPLOY_IAB	None	Inter-cell interference avoidance cannot be enabled in a cell deployed in integrated access and backhaul (IAB) mode.
Low-frequency TDD	gNodeB Adaptive Fast Processing Switch	GNB_ADAPT_F AST_PROCESS_ SW option of the gNodeBParam. ServiceDiffPro cSwitch parameter	None	Adaptive fast processing by the gNodeB cannot be enabled together with inter-cell interference avoidance.
Low-frequency TDD	Any-occasion PDCCH	ANY_OCCASIO N_PDCCH_SW option of the NRDUCellAlgo Switch.LowLat encySwitch parameter	None	Any-occasion PDCCH cannot be enabled together with inter-cell interference avoidance.
Low-frequency TDD	Intra-frequency assistant cell	NRDUCell.Sup plementaryCel lInd set to INTRA_FREQ_A SSISTANT_CEL L	None	Intra-frequency assistant cells do not support inter-cell interference avoidance.
Low-frequency TDD	DL Blanking	NRDUCell.Peer CellSlotAssign ment	<i>Standards Compliance</i>	When inter-cell interference avoidance is enabled, the NRDUCell.Peer CellSlotAssign ment parameter must be set to NOT_CONFIG .

- CSI-RS interference coordination

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	FR1 Max Cell CSI-RS Port Num	NRDUCellCsirs.FR1MaxCellCsirsPortNum	None	CSI-RS interference coordination must be disabled when FR1 Max Cell CSI-RS Port Num is set to 32PORT .
Low-frequency TDD	Interference avoidance during PUSCH/PDSCH scheduling	UL_CELL_LEVEL_INTRF_AVOID and DL_UE_LEVEL_INTRF_AVOID options of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter NRDUCellIntrfIdent.IntrfIdentificationMethod set to UL_DL_MANUAL NRDUCellIntrfBand.InterferenceDirection set to UPLINK_DOWNLINK	<i>LTE and NR Zero Bufferzone</i>	CSI-RS interference coordination cannot be enabled when each of the following conditions is met: - Among the options of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter, only the UL_CELL_LEVEL_INTRF_AVOID and DL_UE_LEVEL_INTRF_AVOID options are selected. - The NRDUCellIntrfIdent.IntrfIdentificationMethod parameter is set to UL_DL_MANUAL. - At least one NRDUCellIntrfBand MO has the NRDUCellIntrfBand.InterferenceDirection parameter set to UPLINK_DOWNLINK. - The BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter is selected.
Low-frequency TDD	Bi-Carrier	BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	None	The BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter is selected.

4.3.2.3 Function Impacts

- Robust weights

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch . <i>CompSwitch</i> parameter	<i>CoMP</i>	Enabling robust weights may decrease the downlink user-perceived rate of downlink coordinated multipoint transmission (DL CoMP) UEs.
Low-frequency TDD	Loose pairing of UEs using robust weights	ROBUST_WT_UE_LOOSE_PAIR_SW option of the NRDUCellDIMimo . <i>DLMuMimoSchSupplementSw</i> parameter	None	Loose pairing of UEs using robust weights must be disabled when the robust weights function is enabled. Otherwise, the benefits of robust weights will be affected.

- Adaptive filtering for SRS channel estimation

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL MU Intrf Reject Precoding Policy	NRDUCellPdsc hPrecode.DIM uIRPrecodePol set to MU_ALLUSER_POWER_ENH	None	Adaptive filtering for SRS channel estimation will yield fewer benefits when it is enabled together with the power-enhancement downlink interference rejection precoding policy for downlink MU-MIMO.
Low-frequency TDD	Enhanced Co-MM	INTRA_GNB_SU_COMM_ENH_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	Adaptive filtering for SRS channel estimation will yield fewer benefits when it is enabled together with enhanced Coordinated Massive MIMO (Co-MM).
Low-frequency TDD	Interference-based subband robust weights	INTRF_BASED_ROBUST_WEIGHT_SW option of the NRDUCellIntrfIdent.ChnMeasCtrlSw parameter	<i>Interference Avoidance</i>	Adaptive filtering for SRS channel estimation will yield fewer benefits when it is enabled together with interference-based subband robust weights.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_LAYER_SW option of the NRDUCellDLRankSelAlgoSw parameter	<i>FWA</i>	If downlink 8-layer transmission for a single FWA user has taken effect, adaptive filtering for SRS channel estimation does not take effect for the user.
Low-frequency TDD	PDSCH self-fusion weights	DL_SELF_FUSION_WEIGHT_SW option of the NRDUCellPdschPrecode.PrecodingAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	The PDSCH self-fusion weights function will yield fewer benefits when it is enabled together with adaptive filtering for SRS channel estimation.

- Rank optimization for cell-edge UEs

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CBF	INTRA_GNB_DL_CS_CBF_SW and INTER_GNB_DL_CS_CBF_SW options of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	Rank optimization for cell-edge UEs does not apply to UEs for which CBF is in effect.
Low-frequency TDD	Fast CBF	QUICK_CBF_SW option of the NRDUCellComp.CsCbfAlgoSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	Rank optimization for cell-edge UEs does not apply to UEs for which fast CBF is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	MU rank boosting	MU_RANK_BOOSTING_SW option of the NRDUCellDimmo.HighLayerMuMimoSw parameter	<i>MIMO (TDD)</i>	Rank optimization for cell-edge UEs does not apply to UEs for which MU rank boosting is in effect.

- Precise MU-MIMO evaluation

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink MU-MIMO layer isolation	DL_MU_MIMO_ISOLATION_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	When downlink MU-MIMO layer isolation is enabled, the benefits of precise MU-MIMO evaluation may change due to changes in UE pairing.
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapBasedFunSw parameter	<i>RedCap</i>	When RedCap UEs are paired for MU-MIMO, the benefits of precise MU-MIMO evaluation slightly decrease because the power usage of MU-MIMO UEs may decrease.

- Dynamic clustering-based grouping

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink MU-MIMO QoS guarantee	DL_MU_MIMO_QOS_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	When dynamic clustering-based grouping takes effect, UEs with low correlations are added to a group first. This affects the downlink experience improvement of downlink MU-MIMO QoS guarantee on home broadband (HBB) users configured with guaranteed rates.

- CCE partial symbol adaptation

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	NR: LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter LTE: LTE_NR_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	When LTE TDD and NR Flash Dynamic Power Sharing takes effect, the benefits of CCE partial symbol adaptation will decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When NR inter-carrier dynamic power sharing takes effect, the benefits of CCE partial symbol adaptation will decrease.
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When NR multi-band power adaptive scheduling takes effect, the benefits of CCE partial symbol adaptation will decrease.
Low-frequency TDD	NR intra-sector power sharing	NR_INTRA_SECTOR_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	None	When NR intra-sector power sharing takes effect, the benefits of CCE partial symbol adaptation will decrease.
Low-frequency TDD	NR PDCCH power sharing	NR_PDCCH_POWER_SHARE_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	None	When NR PDCCH power sharing takes effect, the benefits of CCE partial symbol adaptation will decrease.
Low-frequency TDD	NR DU cell resource management between network slices	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.NetworkSliceAlgorithmSwitch parameter	<i>Network Slicing</i>	When "NR DU cell resource management between network slices" takes effect, the benefits of CCE partial symbol adaptation will decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When Co-MM takes effect, the benefits of CCE partial symbol adaptation will decrease.
Low-frequency TDD	Inter-cell interference avoidance	INTER_CELL_INTERF_AVOID_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When inter-cell interference avoidance takes effect, the benefits of CCE partial symbol adaptation decrease.
Low-frequency TDD	MetaAAU beam adaptation	NRDUCellBeamAlgo.BeamAdaptSwitch	<i>MIMO (TDD)</i>	When MetaAAU beam adaptation takes effect, the benefits of CCE partial symbol adaptation may decrease.
Low-frequency TDD	L/NR Inter-band Deep Coordination	LNR_INTER_BAND_DEEP_COORD_SW option of the NRDUCellResSshrCfg.DynPwrShrAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	When L/NR Inter-band Deep Coordination takes effect, the benefits of CCE partial symbol adaptation will decrease.

- Downlink scheduling based on load and experience

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	MU-MIMO spectral efficiency optimization	SPECTRAL_EFFICIENCY_OPT_SW option of the NRDUCellDLMinmo.DLMuMimoSchSupplementSw parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	When downlink scheduling based on load and experience is enabled but MU-MIMO spectral efficiency optimization is disabled, UEs may be less likely to be paired. If this occurs, the cell throughput decreases.

- Inter-cell interference avoidance

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DRX	BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgorithmSwitch parameter	<i>DRX</i>	When discontinuous reception (DRX) takes effect, inter-cell interference avoidance may not take effect due to failures to determine correlations. If this occurs, the benefits of inter-cell interference avoidance change.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	NR_NR_ANR_SW option of the NRCellAlgoSwitch . AnrSwitch parameter	<i>ANR</i>	When UEs measure cell global identifiers (CGIs), inter-cell interference avoidance may not take effect due to failures to determine correlations. If this occurs, the benefits of inter-cell interference avoidance change.
Low-frequency TDD	Fast CBF	QUICK_CBF_SW option of the NRDUCellComp . CsCbfAlgoSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	Fast CBF does not apply to UEs for which inter-cell interference avoidance is in effect.
Low-frequency TDD	PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch . MuMimoSwitch parameter	<i>MIMO (TDD)</i>	If the PDSCH MU-MIMO pairing rate is high, inter-cell interference avoidance will less likely to take effect because it does not take effect in slots where UEs paired for PDSCH MU-MIMO are scheduled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch . <i>CompSwitch</i> parameter	<i>CoMP</i>	Inter-cell interference avoidance does not apply to UEs for which intra-base-station DL CoMP is in effect.
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	Inter-cell interference avoidance does not apply to UEs for which Co-MM is in effect.
Low-frequency TDD	NR DU cell resource management between network slices	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch . <i>NetworkSliceAlgoSwitch</i> parameter	<i>Network Slicing</i>	Inter-cell interference avoidance is less likely to take effect when it is enabled together with NR DU cell resource management between network slices and the range of available RBs varies among cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOST_BSD_ON_M D_OPT_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	On a given gNodeB, if CA Performance Boost Based on Multi-dimensional Optimization is enabled only for some of the cells for which inter-cell interference avoidance is enabled, the number of cells in which CA Performance Boost Based on Multi-dimensional Optimization takes effect may be less than the number of cells for which this function is enabled. Therefore, it is recommended that CA Performance Boost Based on Multi-dimensional Optimization be enabled for all cells under the gNodeB for which inter-cell interference avoidance is enabled.
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapBasedFunSw parameter	<i>RedCap</i>	Inter-cell interference avoidance does not apply to RedCap UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CCE partial symbol adaptation	CCE_PARTIAL_SYM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When inter-cell interference avoidance takes effect, the benefits of CCE partial symbol adaptation decrease.
Low-frequency TDD	Downlink precise RB estimation	PRECISE_DL_RB_ESTIMATE_SW option of the NRDUCellDISch.DISchAlgoSwitch parameter	<i>Downlink Scheduling</i>	When enabled together with downlink precise RB estimation, interference avoidance may produce increased benefits.
Low-frequency TDD	Fixed AMC Step Value	NRDUCellPdcch.FixedAmcStepValue	None	If Fixed AMC Step Value is set to a value greater than 20 , it is recommended that NRDUCellCollabServ.IntrfAvoidDlTargetIbler be set to the same value as NRDUCellPdcch.DlTargetIbler . If the latter two parameters are set to different values in this situation, inter-cell interference avoidance may decrease the average downlink UE throughput.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Optimized cooperating cell selection	COOP_CELL_ALLOC_OPT_SW option of the NRDUCellComp.CompOptSwitch parameter	<i>CoMP</i>	If optimized cooperating cell selection is enabled together with inter-cell interference avoidance, UEs involved in inter-cell interference avoidance will have more neighboring cells to select from.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Positioning-dedicated SRS transmission in uplink slots	SRS_FOR_POSITION_ON_USL OT_SW option of the NRDUCellPosAlgo.LcsMeasureAlgoSw parameter	<i>Positioning</i>	When both positioning-dedicated SRS transmission in uplink slots and inter-cell interference avoidance are enabled: • If a UE does not support the positioning-dedicated SRS transmission capability, inter-cell interference avoidance cannot take effect for the UE. • If a UE supports the positioning-dedicated SRS transmission capability, inter-cell interference avoidance can take effect for the UE.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

The following base station models support rank optimization for cell-edge UEs, precise MU-MIMO evaluation, target IBLER adjustment in MU-MIMO, downlink

correlation acceleration, downlink scheduling based on load and experience, and CCE partial symbol adaptation:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

The other functions are supported by 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

- Downlink correlation acceleration is supported by all NR TDD-capable low-frequency main control boards and all NR TDD-capable low-frequency UBBPh series baseband processing units (except UBBPh2f and UBBPh3). For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- The other functions are supported by all NR TDD-capable low-frequency main control boards and all NR TDD-capable low-frequency baseband processing units. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable low-frequency RF modules with at least 32T32R can be used. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Cells

The **NRDUCell.FrequencyBand** parameter must be set to **N38**, **N39**, **N40**, **N41**, **N77**, **N78**, or **N79**.

Table 4-1 lists the cell requirements of each iBeam 2.0 function.

Table 4-1 Cell requirements of each iBeam 2.0 function

Function Name	Supported LampSite Cell Type	Supported Low-Frequency TDD Macro Cell Type	Additional Requirement
Robust weights	None	32T/64T cells	The NRDUCell.NrDuCellNetworking-Mode parameter is set to NORMAL_CELL .
Adaptive filtering for SRS channel estimation	None	32T/64T cells	The NRDUCell.NrDuCellNetworking-Mode parameter is set to NORMAL_CELL .

Function Name	Supported LampSite Cell Type	Supported Low-Frequency TDD Macro Cell Type	Additional Requirement
Rank optimization for cell-edge UEs	Distributed massive MIMO cells ^a	32T/64T cells	None
Precise MU-MIMO evaluation	Distributed massive MIMO cells ^a	32T/64T cells	
Target IBLER adjustment in MU-MIMO	Distributed massive MIMO cells ^a	32T/64T cells	
Dynamic clustering-based grouping	None	32T/64T cells	
Downlink correlation acceleration	Distributed massive MIMO cells ^a	32T/64T cells	
CCE partial symbol adaptation	Distributed massive MIMO cells ^a	32T/64T cells	The NRDUCell.SlotAssignment parameter is set to 4_1_DDDSU, 8_2_DDDDDDDSUU, 7_3_DDDSUDDSUU, or 8_2_DDDSUUDDDD. The cell bandwidth must range from 40 MHz to 70 MHz.
Downlink scheduling based on load and experience	Distributed massive MIMO cells ^a	32T/64T cells	None

Function Name	Supported LampSite Cell Type	Supported Low-Frequency TDD Macro Cell Type	Additional Requirement
Inter-cell interference avoidance	None	32T/64T cells	The NRDUCell.NrDuCellNetworking-Mode parameter is set to NORMAL_CELL . A cell with this function enabled and its intra-frequency neighboring cells have the same settings of the NRDUCell.CyclicPrefixLength and NRDUCellCoreset.CommonCtrlResRb-Num parameters.
Precise neighboring cell measurement for SRS IC	None	32T/64T cells	The NRDUCell.NrDuCellNetworking-Mode parameter is set to NORMAL_CELL .
CSI-RS interference coordination	None	32T/64T cells	The NRDUCell.NrDuCellNetworking-Mode parameter is set to NORMAL_CELL or FUSION_CELL .

a: Distributed massive MIMO LampSite cells are those with the **NRDUCell.LampSiteCellFlag** parameter set to **YES** and the **DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter selected.

4.3.5 Others

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

[Table 4-2](#) describes the parameters used for activating iBeam 2.0.

Table 4-2 Parameters used for activating iBeam 2.0

Parameter Name	Parameter ID	Option	Setting Notes
High Precision Beam Phase2 Switch	NRDUCellFeatureSw.HighPrecisionBeamPhase2Sw	None	Set this parameter to ON if iBeam 2.0 is required.

Table 4-3 describes the parameters used for activating weight enhancement.

Table 4-3 Parameters used for activating weight enhancement

Parameter Name	Parameter ID	Option	Setting Notes
Precoding Algorithm Switch	NRDUCellPdschPre-code.PrecodingAlgoSwitch	DL_ROBUS T_WEIGHT_SW	Select this option if PDSCH robust weights are required.
Robust Weight Phase Calculation Method	NRDUCellPdsch.RobustWtPhaseCalcMethod	None	Set this parameter to its recommended value if PDSCH robust weights are required.
SRS Interference Threshold	NRDUCellSrsMeas.SrsIntrfThld	None	Set this parameter to 3 if PDSCH robust weights are required.
SRS Blind Intrf Suppression Degree	NRDUCellSrs.SrsBlindIsDegree	None	Set this parameter to 7 if PDSCH robust weights are required.
SRS Detection Algorithm Switch	NRDUCellSrs.SrsDetectionAlgoSwitch	CHN_EST_A DAPT_FLTR_SW	Select this option if adaptive filtering for SRS channel estimation is required.

Parameter Name	Parameter ID	Option	Setting Notes
Time-domain Channel Correlation Thld	NRDUCellSrsMeas.Ti meChnCorrThld	None	This parameter specifies the time-domain channel correlation threshold used in adaptive filtering for SRS channel estimation. • In fixed wireless access (FWA) scenarios, you are advised to set this parameter to 90. • In other situations, you are advised to set this parameter to 80.

Table 4-4 describes the parameters used for activating scheduling enhancement.

Table 4-4 Parameters used for activating scheduling enhancement

Parameter Name	Parameter ID	Option	Setting Notes
DL MU-MIMO Scheduling Supplement Switch	NRDUCellDlMimo.DL MuMimoSchSupple- mentSw	FAR_UE_RA NK_OPT_S W	Select this option if rank optimization for cell-edge UEs is required.
DL MU-MIMO Scheduling Supplement Switch	NRDUCellDlMimo.DL MuMimoSchSupple- mentSw	PRECISE_M UMIMO_EV AL_SW	Select this option if precise MU-MIMO evaluation is required.
DL MU-MIMO Scheduling Supplement Switch	NRDUCellDlMimo.DL MuMimoSchSupple- mentSw	MUMIMO_I BLER_TARG ET_ADJ_SW	Select this option if target IBLER adjustment in MU-MIMO is required.

Parameter Name	Parameter ID	Option	Setting Notes
MU-MIMO IBLER Target	NRDUCellDlMimo.MuMimoIblerTarget	None	Set this parameter to its recommended value. This parameter takes effect only when the MUMIMO_IBLER_TARGET_ADJ_SW option of the NRDUCellDlMimo.DLMuMimoSchSupplementSw parameter is selected.
DL MU-MIMO Group Mode	NRDUCellPdsch.DlMuMimoGroupMode	None	Set this parameter to DYNAMIC_CLUSTER_GROUP if dynamic clustering-based grouping is required.
DL MU-MIMO Scheduling Supplement Switch	NRDUCellDlMimo.DLMuMimoSchSupplementSw	DL_CORR_ACCELERATION_SW	Select this option if downlink correlation acceleration is required.
DL MU-MIMO Correlation Accelerator Mode	NRDUCellDlMimo.DLMuCorrAcceleratorMode	None	Set this parameter to its recommended value.
Downlink Schedule Algorithm Switch	NRDUCellDlSch.DlSchAlgoSwitch	TAIL_PKT_MCS_OPT_SW	You are advised to select this option to achieve greater benefits when any of the following functions is required: precise MU-MIMO evaluation, target IBLER adjustment in MU-MIMO, and dynamic clustering-based grouping.
PDCCH Algorithm Switch	NRDUCellPdcch.PdcchAlgoSwitch	CCE_PARTIAL_SYM_ADAPT_SW	Select this option if CCE partial symbol adaptation is required.

Parameter Name	Parameter ID	Option	Setting Notes
PDCCH Algorithm Switch	NRDUCellPdcch.<i>PdchhAlgoSwitch</i>	PDCCH_BLI ND_DET_A SSIGN_OPT _SW	You are advised to select this option if CCE partial symbol adaptation is required. The purpose is to balance blind detection times across different PDCCH aggregation levels, so as to prevent uplink and downlink CCE allocation failures due to insufficient blind detection times.
Blacklist Control Extension Switch	gNBUECompatSw.<i>BlacklistCtrlExtSwitch</i>	PDCCH_SY M_DYNAMI C_ALLOC_S W_OFF	Select this option if there are UEs incompatible with CCE partial symbol adaptation on the network.
Downlink Schedule Algorithm Switch	NRDUCellDISch.<i>DISchhAlgoSwitch</i>	LOAD_BAS ED_DL_EXP _SCH_SW	Select this option if downlink scheduling based on load and experience is required.
DL Adaptive Scheduling PRB Threshold	NRDUCellDISch.<i>DLAdaptSchPrbThld</i>	None	This parameter specifies the per-slot PRB usage threshold used in downlink scheduling based on load and experience. Set this parameter to its recommended value.

Table 4-5 describes the parameters used for activating interference-aware scheduling enhancement.

Table 4-5 Parameters used for activating interference-aware scheduling enhancement

Parameter Name	Parameter ID	Option	Setting Notes
Multi-Cell MIMO Switch	NRDUCellCollabServ. <i>MultiCellMimoSwitch</i>	INTER_CELL_INTRF_AVOID_SW	Select this option if inter-cell interference avoidance is required.
Interference Avoid Downlink Target IBLER	NRDUCellCollabServ. <i>IntrfAvoidDLTargetIbler</i>	None	This parameter specifies the downlink target IBLER for UEs involved in inter-cell interference avoidance. This parameter takes effect only when the INTER_CELL_INTRF_AVOID_SW option of the NRDUCellCollabServ. <i>MultiCellMimoSwitch</i> parameter is selected. Retain the default value.
SRS Detection Algorithm Switch	NRDUCellSrs. <i>SrsDetectionAlgoSwitch</i>	SRS_IC_NC_ELL_PRECISE_MEAS_SW	Select this option if precise neighboring cell measurement for SRS IC is required.
CSI Switch	NRDUCellCsirs. <i>CsiSwitch</i>	CSIRS_INTRF_COORD_SW	Select this option if CSI-RS interference coordination is required.

4.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

To enable iBeam 2.0:

```
//Enabling iBeam 2.0  
MOD NRDUCELLFEATURESW: NrDuCellId=0, HighPrecisionBeamPhase2Sw=ON;
```

To enable functions related to weight enhancement:

```
//Enabling PDSCH robust weights  
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, PrecodingAlgoSwitch=DL_ROBUST_WEIGHT_SW-1;  
//Setting the SRS polarization phase for PDSCH robust weights  
MOD NRDUCELLPDSCH: NrDuCellId=0, RobustWtPhaseCalcMethod=SRS_H_BASED_CALCULATION;  
//(Optional) Setting the SRS interference threshold for PDSCH robust weights  
MOD NRDUCELLSRSMEAS: NrDuCellId=0, SrsIntrfThld=3;  
//(Optional) Setting the blind SRS interference suppression degree for PDSCH robust weights  
MOD NRDUCELLSRS: NrDuCellId=0, SrsBlindIsDegree=7;  
  
//Enabling adaptive filtering for SRS channel estimation  
MOD NRDUCELLSRS: NrDuCellId=0, SrsDetectionAlgoSwitch=CHN_EST_ADAPT_FLTR_SW-1;  
MOD NRDUCELLSRSMEAS: NrDuCellId=0, TimeChnCorrThld=80;
```

To enable functions related to scheduling enhancement:

```
//Enabling rank optimization for cell-edge UEs  
MOD NRDUCELLDLMIMO: NrDuCellId=0, DLMuMimoSchSupplementSw=FAR_UE_RANK_OPT_SW-1;  
  
//Enabling precise MU-MIMO evaluation  
MOD NRDUCELLDLMIMO: NrDuCellId=0, DLMuMimoSchSupplementSw=PRECISE_MUMIMO_EVAL_SW-1;  
  
//Enabling target IBLER adjustment in MU-MIMO  
MOD NRDUCELLDLMIMO: NrDuCellId=0,  
DLMuMimoSchSupplementSw=MUMIMO_IBLER_TARGET_ADJ_SW-1;  
//Adjusting the target IBLER for MU-MIMO  
MOD NRDUCELLDLMIMO: NrDuCellId=0, MuMimoIblerTarget=15;  
  
//Enabling dynamic clustering-based grouping  
MOD NRDUCELLPDSCH: NrDuCellId=0, DLMuMimoGroupMode=DYNAMIC_CLUSTER_GROUP;  
  
//Enabling downlink correlation acceleration  
MOD NRDUCELLDLMIMO: NrDuCellId=0, DLMuMimoSchSupplementSw=DL_CORR_ACCELERATION_SW-1;  
//(Optional) Configuring the accelerator mode for calculating correlations used in downlink MU-MIMO  
MOD NRDUCELLDLMIMO: NrDuCellId=0, DLMuCorrAcceleratorMode=PRECALCULATION;  
  
//(Optional, recommended when precise MU-MIMO evaluation, target IBLER adjustment in MU-MIMO, or  
dynamic clustering-based grouping is enabled) Enabling MCS optimization for tail packets  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=TAIL_PKT_MCS_OPT_SW-1;  
  
//Enabling CCE partial symbol adaptation  
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=CCE_PARTIAL_SYM_ADAPT_SW-1;  
//(Recommended when CCE partial symbol adaptation is enabled) Enabling PDCCH blind detection  
assignment optimization, which aims to balance blind detection times across different PDCCH aggregation  
levels, so as to prevent uplink and downlink CCE allocation failures due to insufficient blind detection times  
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_BLIND_DET_ASSIGN_OPT_SW-1;  
//Enabling blacklist control for dynamic PDCCH symbol allocation if there are UEs incompatible with CCE  
partial symbol adaptation on the network (using the UeFeatureValueContent parameter set to  
ec7fee769ff7e0fff0 as an example)  
ADD GNBUEINFO: UeInfoIndex=1, UeInfoType=UE_FEATUREVALUE,  
UeFeatureValueContent="ec7fee769ff7e0fff0",  
UeFeatureValueMask="FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF", UeFeatureValueVersion=VERSION1;  
ADD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,  
BlacklistCtrlExtSwitch=PDCCH_SYM_DYNAMIC_ALLOC_SW_OFF-1;  
  
//Enabling downlink scheduling based on load and experience  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=LOAD_BASED_DL_EXP_SCH_SW-1;  
//Setting the PRB usage threshold for adaptive downlink scheduling  
MOD NRDUCELLDLSCH: NrDuCellId=0, DAdaptSchPrbThld=95;
```

To enable functions related to interference-aware scheduling enhancement:

```
//Enabling inter-cell interference avoidance  
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, MultiCellMimoSwitch=INTER_CELL_INTRF_AVOID_SW-1,  
IntrfAvoidDlTargetIbler=20;
```



```
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=PRECISE_DL_RB_ESTIMATE_SW-1;  
//Enabling precise neighboring cell measurement for SRS IC  
MOD NRDUCELLSRS: NrDuCellId=0, SrsDetectionAlgoSwitch=SRS_IC_NCELL_PRECISE_MEAS_SW-1;  
//Enabling CSI-RS interference coordination  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=CSIRS_INTRF_COORD_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

To disable functions related to weight enhancement:

```
//Disabling PDSCH robust weights  
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, PrecodingAlgoSwitch=DL_ROBUST_WEIGHT_SW-0;  
  
//Disabling adaptive filtering for SRS channel estimation  
MOD NRDUCELLSRS: NrDuCellId=0, SrsDetectionAlgoSwitch=CHN_EST_ADAPT_FLTR_SW-0;
```

To disable functions related to scheduling enhancement:

```
//Disabling rank optimization for cell-edge UEs  
MOD NRDUCELLDLMIMO: NrDuCellId=0, DLMuMimoSchSupplementSw=FAR_UE_RANK_OPT_SW-0;  
  
//(Optional) Disabling MCS optimization for tail packets  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=TAIL_PKT_MCS_OPT_SW-0;  
  
//Disabling precise MU-MIMO evaluation  
MOD NRDUCELLDLMIMO: NrDuCellId=0, DLMuMimoSchSupplementSw=PRECISE_MUMIMO_EVAL_SW-0;  
  
//Disabling target IBLER adjustment in MU-MIMO  
MOD NRDUCELLDLMIMO: NrDuCellId=0,  
DLMuMimoSchSupplementSw=MUMIMO_IBLER_TARGET_ADJ_SW-0;  
  
//Disabling downlink correlation acceleration  
MOD NRDUCELLDLMIMO: NrDuCellId=0, DLMuMimoSchSupplementSw=DL_CORR_ACCELERATION_SW-0;  
  
//Disabling blacklist control for dynamic PDCCH symbol allocation  
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,  
BlacklistCtrlExtSwitch=PDCCH_SYM_DYNAMIC_ALLOC_SW-OFF-0;  
//Disabling CCE partial symbol adaptation  
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=CCE_PARTIAL_SYM_ADAPT_SW-0;  
//Disabling PDCCH blind detection assignment optimization  
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_BLIND_DET_ASSIGN_OPT_SW-0;  
  
//Disabling downlink scheduling based on load and experience  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=LOAD_BASED_DL_EXP_SCH_SW-0;
```

To disable functions related to interference-aware scheduling enhancement:

```
//Disabling inter-cell interference avoidance  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=PRECISE_DL_RB_ESTIMATE_SW-0;  
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, MultiCellMimoSwitch=INTER_CELL_INTRF_AVOID_SW-0;  
  
//Disabling precise neighboring cell measurement for SRS IC  
MOD NRDUCELLSRS: NrDuCellId=0, SrsDetectionAlgoSwitch=SRS_IC_NCELL_PRECISE_MEAS_SW-0;  
  
//Disabling CSI-RS interference coordination  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=CSIRS_INTRF_COORD_SW-0;
```

To disable iBeam 2.0:

```
//Disabling iBeam 2.0  
MOD NRDUCELLFEATURESW: NrDuCellId=0, HighPrecisionBeamPhase2Sw=OFF;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Observe the **User Downlink Average Throughput (DU)** before and after iBeam 2.0 is enabled. If the value of this indicator increases, iBeam 2.0 has taken effect.

4.4.3 Network Monitoring

After iBeam 2.0 is enabled, use the performance indicators described in [4.2 Network Analysis](#) to monitor cell performance.

5 iBeam 3.0

5.1 Principles

iBeam 3.0 has been introduced to further increase the downlink user-perceived rate in medium- and heavy-load enhanced Mobile Broadband (eMBB) cells that offer contiguous coverage. This feature applies to TDD massive MIMO cells. It improves downlink performance and increases the downlink user-perceived rate through weight enhancement, scheduling enhancement, and interference suppression enhancement.

To enable iBeam 3.0 and its various functions for these enhancements, the **NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw** parameter must be set to ON.

5.1.1 Weight Enhancement

PDSCH Self-Fusion Weights

A low signal-to-noise ratio (SNR) leads to poor SRS quality, inaccurate SRS channel estimation, and ultimately inaccurate weights. PDSCH self-fusion weights can now be used. They are generated by combining statistical SRS-based weights with instantaneous channel characteristics. These self-fusion weights are more accurate and thereby increase the downlink user-perceived rate.

This function is controlled by the **DL_SELF_FUSION_WEIGHT_SW** option of the **NRDUCellPdschPrecode.PrecodingAlgoSwitch** parameter. PDSCH self-fusion weights take effect only when both the SRS PreSINR and time-domain channel correlation meet respective thresholds. The thresholds are configured and used as follows:

- The SRS PreSINR threshold for PDSCH self-fusion weights is specified by the **NRDUCellPdschPrecode.SelfFusionWtSrsPresinrThld** parameter. When the SRS PreSINR of a UE is less than this threshold, PDSCH self-fusion weights can take effect for this UE. A higher threshold results in a larger number of UEs for which PDSCH self-fusion weights take effect and a higher upper limit for the increase in the downlink user-perceived rate. A lower threshold results in the opposite effects. If there are a large number of UEs (for example, when

the average number of RRC_CONNECTED UEs is greater than 200), this SRS PreSINR threshold can be increased to raise the upper limit for the function benefits. If there are a small number of UEs (for example, when the average number of RRC_CONNECTED UEs is less than 100), setting this threshold to an excessively large value may lower the upper limit for the function benefits.

- The time-domain channel correlation threshold for PDSCH self-fusion weights is specified by the **NRDUCellPdschPrecode.SelfFusionWtRhoThld** parameter. When the time-domain channel correlation of a UE is less than this threshold, PDSCH self-fusion weights can take effect for this UE. A higher threshold results in a larger number of UEs for which PDSCH self-fusion weights take effect and a larger increase in the downlink user-perceived rate. A lower threshold results in the opposite effects.

When the PDSCH self-fusion weights function is enabled, it is recommended that downlink rank adaptation for cell edge UEs performing small-packet services also be enabled for greater benefits. (Downlink rank adaptation for cell edge UEs performing small-packet services is controlled by the **DL_SMALL_PKT_CEU_RANK_ADAPT_SW** option of the **NRDUCellDLRank.DlRankSelAlgoSw** parameter.)

PDCCH Robust Weights

The PDCCH typically uses discrete Fourier transform (DFT) weights to generate static beams, which have poor directional accuracy. SRS-based weights can now be used to generate robust weights. By using the more accurate robust weights instead of DFT weights, the PDCCH can have its beams targeted at UEs more accurately, increasing the downlink user-perceived rate.

The PDCCH robust weights function is controlled by the **PDCCH_ROBUST_WEIGHT_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter.

This function uses an SRS PreSINR threshold, which is specified by the **NRDUCellPdcchAlgo.PdcchRobWtSrsPresinrThld** parameter. When the SRS PreSINR of a UE is in the range of **(NRDUCellPdcchAlgo.PdcchRobWtSrsPresinrThld, NRDUCellPdschPrecode.SelfFusionWtSrsPresinrThld)**, PDCCH robust weights can take effect for this UE. A lower threshold results in a higher proportion of times that PDCCH robust weights take effect and a larger increase in the downlink user-perceived rate. However, if this threshold is set to an excessively small value, the SRS measurement accuracy is reduced, and the downlink user-perceived rate decreases. A higher threshold results in a lower proportion of times that PDCCH robust weights take effect and a smaller increase in the downlink user-perceived rate.

This function also uses a conversion coefficient to calculate the beamforming gain. This coefficient is specified by the **NRDUCellPdcchAlgo.PdcchRobWtBFGainCoeff** parameter. A larger value of this parameter results in a larger beamforming gain of PDCCH robust weights, a more significant decrease in the average PDCCH aggregation level, and a larger increase in the downlink UE throughput. However, if the aggregation level is reduced excessively, the probability of PDCCH miss-detection increases, which may decrease the downlink UE traffic volume. A smaller value of this parameter results in a smaller beamforming gain of PDCCH robust weights and a smaller increase in the downlink UE throughput.

5.1.2 Scheduling Enhancement

SINR Optimization for MCS Selection

SINR optimization for MCS selection adjusts the SINRs used to select MCSs for MU-MIMO UEs, thereby enhancing UEs' spectral efficiency and improving the downlink user-perceived rate.

SINR optimization for MCS selection is controlled by the **DL_SMART_AMC_SW** option of the **NRDUCellDlMimo.DLMuMimoSchSupplementSw** parameter.

Downlink Rank Robustness

Downlink rank robustness enables the base station to check filtered statistics to determine whether to use instantaneous UE-reported ranks or filtered-statistics-based ranks. This function improves the accuracy of MCS selection and the downlink user-perceived rate.

Downlink rank robustness is controlled by the **ROBUST_RANK_SW** option of the **NRDUCellDlAmc.DlMcsSelAlgoSw** parameter. If this option is deselected, the base station uses UE-reported ranks by default.

When downlink rank robustness is enabled, it is recommended that the following functions also be enabled for greater benefits: CSI reporting period adaptation (controlled by the **CSI_REPORT_PERIOD_ADAPT_SWITCH** option of the **NRDUCellPucch.CsiResoureAlgoSwitch** parameter) and PUCCH RB adaptation (controlled by the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter).

"Rank robustness optimization in mobility scenarios" is recommended for cells serving high-speed UEs. It improves the rank accuracy for mobile UEs, whose CSI-RS beams are subject to changes. This function enables the base station to refilter the rank reported by a UE after the CSI-RS beams of the UE change. "Rank robustness optimization in mobility scenarios" is controlled by the **ROBUST_RANK_MOBILITY_OPT_SW** option of the **NRDUCellDlAmc.DlMcsSelAlgoSw** parameter.

Downlink Small Packet Shaping

Given the significant variation in traffic models among UEs, downlink small packet shaping has been introduced to balance resource allocation in the time domain. This function adaptively selects UEs for scheduling based on the RB resources occupied by them, improving the user-perceived rate.

Downlink small packet shaping is controlled by the **DL_SMALL_PACKET_SHAPE_SCH_SW** option of the **NRDUCellDlSch.DlSchAlgoSwitch** parameter.

Power Optimization for Downlink Retransmissions

In a hybrid automatic repeat request (HARQ) process during downlink scheduling for retransmissions, a received error data packet is stored in a HARQ buffer and combined with the subsequently received retransmitted data packet for more reliable decoding. This process consumes much power. Power optimization for

downlink retransmissions has therefore been introduced to address this issue. This function reduces the power used for retransmissions and reallocates the freed-up power to initial transmissions, further improving spectral efficiency.

Power optimization for downlink retransmissions is controlled by the **DL_RETRANS_PWR_OPT_SW** option of the **NRDUCellChnPwr.ChnPwrAlgoSwitch** parameter.

Downlink Retransmission Scheduling Priority Adjustment

Downlink retransmission scheduling priority adjustment adjusts the scheduling priorities of UEs involved in retransmissions based on user experience, thereby improving the downlink user-perceived rate.

Downlink retransmission scheduling priority adjustment is controlled by the **DL_RETRANS_PRI_ADJ_SW** option of the **NRDUCellDISch.DISchAlgoSwitch** parameter.

When downlink retransmission scheduling priority adjustment is enabled, it is recommended that resource-based adaptive scheduling for massive MIMO also be enabled for greater benefits. (Resource-based adaptive scheduling for massive MIMO is controlled by the **RES_BASED_MM_ADAPT_SCH_SW** option of the **NRDUCellServExp.ServiceExpAlgoSwitch** parameter.)

Downlink Slot Awareness Scheduling Optimization

To account for both the difference in the number of symbols between slots and the fluctuating nature of UEs' incoming services, downlink slot awareness scheduling optimization has been introduced. This function schedules UEs having large buffer sizes into slots with more symbols, thereby improving the downlink user-perceived rate.

Downlink slot awareness scheduling optimization is controlled by the **DL_SLOT_AWARE_SCH_OPT_SW** option of the **NRDUCellDISch.DISchAlgoSwitch** parameter.

Intelligent MCS Optimization for Downlink MU-MIMO

Intelligent MCS optimization for downlink MU-MIMO accurately calculates the MCSs for MU-MIMO UEs, thereby enhancing UEs' spectral efficiency and improving the downlink user-perceived rate.

Intelligent MCS optimization for downlink MU-MIMO is controlled by the **DL_MU_MCS_INTEL_OPT_SW** option of the **NRDUCellAiAlgo.AiAmcAlgoSwitch** parameter.

5.1.3 Interference Suppression Enhancement

In contiguous networking, SRS interference changes rapidly and constantly. Blind SRS interference suppression enhancement uses the differences between interference signals and wanted signals to reduce or eliminate interference for UEs severely affected by SRS interference. This function improves the accuracy of SRS measurement and thereby increases the average downlink UE throughput.

Blind SRS interference suppression enhancement is controlled by the **SRS_ENH_BLIND_IS_SW** option of the **NRDUCellSrs.SrsDetectionAlgoSwitch** parameter.

5.2 Network Analysis

5.2.1 Benefits

iBeam 3.0 increases the **User Downlink Average Throughput (DU)**. The actual benefits of iBeam 3.0 depend on factors such as the traffic model, network load, and PCI planning. This feature is recommended for intra-frequency contiguous massive MIMO networks when the following conditions are met:

- The downlink PRB usage (equal to $\frac{\text{N.PR.B.DL.Used.Avg}}{\text{N.PR.B.DL.Avail.Avg}}$) ranges from 30% to 70%.
- The proportion of SRS PreSINRs less than 0 dB is greater than 50%. This proportion equals $\frac{(\text{N.U.L.S.R.S.PreSINR.Index0} + \text{N.U.L.S.R.S.PreSINR.Index1} + \text{N.U.L.S.R.S.PreSINR.Index2} + \text{N.U.L.S.R.S.PreSINR.Index3})}{(\text{N.U.L.S.R.S.PreSINR.Index0} + \text{N.U.L.S.R.S.PreSINR.Index1} + \text{N.U.L.S.R.S.PreSINR.Index2} + \text{N.U.L.S.R.S.PreSINR.Index3} + \text{N.U.L.S.R.S.PreSINR.Index4} + \text{N.U.L.S.R.S.PreSINR.Index5})}$.
- The downlink CCE allocation failure rate is greater than 5%. This rate equals $\frac{(\text{N.C.C.E.DL.AllocReq.Num} - \text{N.C.C.E.DL.AggLvl1Num} - \text{N.C.C.E.DL.AggLvl2Num} - \text{N.C.C.E.DL.AggLvl4Num} - \text{N.C.C.E.DL.AggLvl8Num} - \text{N.C.C.E.DL.AggLvl16Num})}{\text{N.C.C.E.DL.AllocReq.Num}}$.

For other networks or networks that serve UEs moving at high speed (for example, on highways), some functions offer fewer or even no benefits. In addition, iBeam 3.0 offer fewer benefits if any of the functions for weight enhancement, scheduling enhancement, and interference suppression enhancement is disabled.

PDCCH robust weights offer fewer or no benefits if the average uplink CCE aggregation level is less than 3. The average uplink CCE aggregation level equals $\frac{(1 \times \text{N.C.C.E.U.L.AggLvl1Num} + 2 \times \text{N.C.C.E.U.L.AggLvl2Num} + 4 \times \text{N.C.C.E.U.L.AggLvl4Num} + 8 \times \text{N.C.C.E.U.L.AggLvl8Num} + 16 \times \text{N.C.C.E.U.L.AggLvl16Num})}{(\text{N.C.C.E.U.L.AggLvl1Num} + \text{N.C.C.E.U.L.AggLvl2Num} + \text{N.C.C.E.U.L.AggLvl4Num} + \text{N.C.C.E.U.L.AggLvl8Num} + \text{N.C.C.E.U.L.AggLvl16Num})}$.

When the PDSCH self-fusion weights function is enabled, it is recommended that downlink rank adaptation for cell edge UEs performing small-packet services also be enabled for greater benefits. (Downlink rank adaptation for cell edge UEs performing small-packet services is controlled by the **DL_SMALL_PKT_CEU_RANK_ADAPT_SW** option of the **NRDUCellDLRank.DlRankSelAlgoSw** parameter.)

When downlink rank robustness is enabled, it is recommended that the following functions also be enabled for greater benefits: CSI reporting period adaptation (controlled by the **CSI_REPORT_PERIOD_ADAPT_SWITCH** option of the **NRDUCellPucch.CsiResoureAlgoSwitch** parameter) and PUCCH RB adaptation (controlled by the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter).

When downlink retransmission scheduling priority adjustment is enabled, it is recommended that resource-based adaptive scheduling for massive MIMO also be enabled for greater benefits. (Resource-based adaptive scheduling for massive MIMO is controlled by the **RES_BASED_MM_ADAPT_SCH_SW** option of the **NRDUCellServExp.ServiceExpAlgoSwitch** parameter.)

Blind SRS interference suppression enhancement may offer fewer benefits in contiguous networking where UEs exhibit concentrated distribution within the cells.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.2.2 Impacts](#).

iBeam 3.0 may have the following impacts on the network:

- The downlink MCS index may increase.
- The MU-MIMO pairing rate may increase. For details about the impacts of the increase on related KPIs, see the "Impacts" section of MU-MIMO (low-frequency TDD) in *MIMO (TDD)*. (MU-MIMO pairing rate = $\frac{\text{N.ChMeas.MIMO.DL.Pair.PRB}}{(\text{N.PRB.DL.Used.Avg} \times \text{N.DL.PDSCH.Tti.Num})}$)
- The average number of layers paired for downlink MIMO (indicated by the **N.ChMeas.MIMO.DL.Pair.Layer.Avg** counter) may change.
- The downlink PRB usage may decrease. (Downlink PRB usage = $\frac{\text{N.PRB.DL.Used.Avg}}{\text{N.PRB.DL.Avail.Avg}}$)
- The values of the following scheduling-related indicators may change:
 - Downlink average rank, which equals $(1 \times \text{N.PDSCH.InitTbDL.Rank1} + 2 \times \text{N.PDSCH.InitTbDL.Rank2} + 3 \times \text{N.PDSCH.InitTbDL.Rank3} + 4 \times \text{N.PDSCH.InitTbDL.Rank4}) / (\text{N.PDSCH.InitTbDL.Rank1} + \text{N.PDSCH.InitTbDL.Rank2} + \text{N.PDSCH.InitTbDL.Rank3} + \text{N.PDSCH.InitTbDL.Rank4})$
 - Downlink IBLER, which equals $(\text{N.DL.SCH.QPSK.ErrTB.Ibler} + \text{N.DL.SCH.16QAM.ErrTB.Ibler} + \text{N.DL.SCH.64QAM.ErrTB.Ibler} + \text{N.DL.SCH.256QAM.ErrTB.Ibler} + \text{N.DL.SCH.1024QAM.ErrTB.Ibler}) / (\text{N.DL.SCH.QPSK.TB} + \text{N.DL.SCH.16QAM.TB} + \text{N.DL.SCH.64QAM.TB} + \text{N.DL.SCH.256QAM.TB} + \text{N.DL.SCH.1024QAM.TB})$
 - Downlink RBLER, which equals $(\text{N.DL.SCH.QPSK.ErrTB.Rbler} + \text{N.DL.SCH.16QAM.ErrTB.Rbler} + \text{N.DL.SCH.64QAM.ErrTB.Rbler} + \text{N.DL.SCH.256QAM.ErrTB.Rbler} + \text{N.DL.SCH.1024QAM.ErrTB.Rbler}) / (\text{N.DL.SCH.QPSK.TB} + \text{N.DL.SCH.16QAM.TB} + \text{N.DL.SCH.64QAM.TB} + \text{N.DL.SCH.256QAM.TB} + \text{N.DL.SCH.1024QAM.TB})$
- The equivalent PDCCH CCE usage of a cell may change. (Equivalent PDCCH CCE usage of a cell = $(\text{N.CCE.ULDCI.Used.Equivalent} + \text{N.CCE.DLDCI.Used.Equivalent}) / \text{N.CCE.Avail.Equivalent}$)
- The average CCE aggregation level and CCE allocation failure rate may decrease for both the uplink and downlink.
 - Average uplink CCE aggregation level = $(1 \times \text{N.CCE.UL.AggLvl1Num} + 2 \times \text{N.CCE.UL.AggLvl2Num} + 4 \times \text{N.CCE.UL.AggLvl4Num} + 8 \times \text{N.CCE.UL.AggLvl8Num} + 16 \times \text{N.CCE.UL.AggLvl16Num}) / (\text{N.CCE.UL.AggLvl1Num} + \text{N.CCE.UL.AggLvl2Num} +$

- $$\text{N.CCE.Ul.AggLvl4Num} + \text{N.CCE.Ul.AggLvl8Num} + \text{N.CCE.Ul.AggLvl16Num})$$
- Average downlink CCE aggregation level = $(1 \times \text{N.CCE.DL.AggLvl1Num} + 2 \times \text{N.CCE.DL.AggLvl2Num} + 4 \times \text{N.CCE.DL.AggLvl4Num} + 8 \times \text{N.CCE.DL.AggLvl8Num} + 16 \times \text{N.CCE.DL.AggLvl16Num}) / (\text{N.CCE.DL.AggLvl1Num} + \text{N.CCE.DL.AggLvl2Num} + \text{N.CCE.DL.AggLvl4Num} + \text{N.CCE.DL.AggLvl8Num} + \text{N.CCE.DL.AggLvl16Num})$
 - Uplink CCE allocation failure rate = $(\text{N.CCE.Ul.AllocReq.Num} - \text{N.CCE.Ul.AggLvl1Num} - \text{N.CCE.Ul.AggLvl2Num} - \text{N.CCE.Ul.AggLvl4Num} - \text{N.CCE.Ul.AggLvl8Num} - \text{N.CCE.Ul.AggLvl16Num}) / \text{N.CCE.Ul.AllocReq.Num}$
 - Downlink CCE allocation failure rate = $(\text{N.CCE.DL.AllocReq.Num} - \text{N.CCE.DL.AggLvl1Num} - \text{N.CCE.DL.AggLvl2Num} - \text{N.CCE.DL.AggLvl4Num} - \text{N.CCE.DL.AggLvl8Num} - \text{N.CCE.DL.AggLvl16Num}) / \text{N.CCE.DL.AllocReq.Num}$
 - The first packet delay (indicated by the **N.Traffic.DL.RlcFirstPktDelay.Time** counter) may change.
 - The duration in which BWP2 is in effect (indicated by the **N.Bwp2.Dur.Total** counter) may increase.
 - In light- and medium-load cells where tail packets account for an excessively large proportion of data volume, the **Cell Downlink Average Throughput (DU)** may increase but the **User Downlink Average Throughput (DU)** may decrease. The data volume proportion of tail packets equals $\text{N.ThpVol.DL.LastSlot} / \text{N.ThpVol.DL}$.

5.3 Requirements

5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-100200	iBeam 3.0	NR0S00BEAM30	Per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

- PDSCH self-fusion weights

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	PDSCH self-fusion weights can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .
Low-frequency TDD	Low-speed cell	NRDUCell.HighSpeedFlag set to LOW_SPEED	None	PDSCH self-fusion weights can be enabled only for low-speed cells.

- PDCCH robust weights

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	PDCCH robust weights can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .
Low-frequency TDD	PDSCH self-fusion weights	DL_SELF_FUSION_WEIGHT_SW option of the NRDUCellPdschPrecode.PrecodingAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	PDCCH robust weights can be enabled only when the PDSCH self-fusion weights function is enabled.

- SINR optimization for MCS selection

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	SINR optimization for MCS selection can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .
Low-frequency TDD	Downlink MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	SINR optimization for MCS selection can take effect only when downlink MU-MIMO is enabled.

- Downlink rank robustness

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Downlink rank robustness can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .

- Rank robustness optimization in mobility scenarios

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink rank robustness	ROBUST_RANK_SW option of the NRDUCellIDIA mc.DIMcsSelAI goSw parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	"Rank robustness optimization in mobility scenarios" (controlled by the ROBUST_RANK_MOBILITY_OPT_SW option of the NRDUCellIDIA mc.DIMcsSelAI goSw parameter) can be enabled only when downlink rank robustness is enabled.

- Downlink small packet shaping

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Downlink small packet shaping can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .

- Power optimization for downlink retransmissions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Power optimization for downlink retransmissions can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .

- Downlink retransmission scheduling priority adjustment

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Downlink retransmission scheduling priority adjustment can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .

- Downlink slot awareness scheduling optimization

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Downlink slot awareness scheduling optimization can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .

- Intelligent MCS optimization for downlink MU-MIMO

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Intelligent MCS optimization for downlink MU-MIMO can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .
Low-frequency TDD	Downlink MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Intelligent MCS optimization for downlink MU-MIMO can take effect only when downlink MU-MIMO is enabled.

- Blind SRS interference suppression enhancement

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Blind SRS interference suppression enhancement takes effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	None	Blind SRS interference suppression enhancement can be enabled only when UE-characteristic-based SRS period adaptation is enabled.

5.3.2.2 Mutually Exclusive Functions

- PDCCH robust weights

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Tridimensional coverage mode	NRDUCell.TridCommMode set to TRID_COVERAGE_MODE	None	When the NRDUCell.TridCommMode parameter is set to TRID_COVERAGE_MODE , PDCCH robust weights cannot be enabled.

- Blind SRS interference suppression enhancement

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	When blind SRS interference suppression enhancement is enabled, the NRDUCell.HighSpeedFlag parameter cannot be set to HIGH_SPEED .

5.3.2.3 Function Impacts

- PDSCH self-fusion weights

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Adaptive filtering for SRS channel estimation	CHN_EST_ADAPT_FLTR_SW option of the NRDUCellSrs.SrsDetectionAlgorithmSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	The PDSCH self-fusion weights function will yield fewer benefits when it is enabled together with adaptive filtering for SRS channel estimation.
Low-frequency TDD	Enhanced Co-MM	INTRA_GNB_SU_COMM_ENH_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	The PDSCH self-fusion weights function will yield fewer benefits when it is enabled together with enhanced Co-MM.
Low-frequency TDD	Interference-based subband robust weights	INTRF_BASED_ROBUST_WEIGHT_SW option of the NRDUCellIntrflident.ChnMeasCtrlSw parameter	<i>Interference Avoidance</i>	The PDSCH self-fusion weights function will yield fewer benefits when it is enabled together with interference-based subband robust weights.

- PDCCH robust weights

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast intra-gNodeB CA	CA_ULTRA_LOW_LATENCY_SPLIT_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoExtSwitch parameter	<i>CA Experience Improvement</i>	The PDCCH robust weights function will yield fewer benefits when it is enabled together with fast intra-gNodeB CA.
Low-frequency TDD	Fast inter-gNodeB CA	INTER_GNB_ULTRALOW_LAT_SPLIT_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoExtSwitch parameter	<i>CA Experience Improvement</i>	The PDCCH robust weights function will yield fewer benefits when it is enabled together with fast inter-gNodeB CA.

- Power optimization for downlink retransmissions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Precise RB control for downlink retransmissions	DL_RETRANS_PRECISE_RB_CTRL_SW option of the NRDUCellDisch.DISchAlgoSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	Precise RB control for downlink retransmissions preferentially takes effect when it is enabled together with power optimization for downlink retransmissions. Power optimization for downlink retransmissions can take effect only when precise RB control for downlink retransmissions is not in effect.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

SINR optimization for MCS selection, downlink rank robustness, downlink small packet shaping, downlink retransmission scheduling priority adjustment, downlink slot awareness scheduling optimization, and intelligent MCS optimization for downlink MU-MIMO are supported by the following base station models:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

The other functions are supported by 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR TDD-capable low-frequency main control boards and NR TDD-capable low-frequency UBBPg and later series baseband processing units can be used. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable low-frequency RF modules with at least 32T32R can be used. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Cells

The **NRDUCell.FrequencyBand** parameter must be set to **N38**, **N39**, **N40**, **N41**, **N77**, **N78**, or **N79**.

Table 5-1 lists the cell requirements of each iBeam 3.0 function.

Table 5-1 Cell requirements of each iBeam 3.0 function

Function Name	Supported LampSite Cell Type	Supported Low-Frequency TDD Macro Cell Type	Additional Requirement
PDSCH self-fusion weights	None	32T/64T cells	The NRDUCell.NrDuCellNetworking-Mode parameter is set to NORMAL_CELL .
PDCCH robust weights	None	32T/64T cells	The NRDUCell.NrDuCellNetworking-Mode parameter is set to NORMAL_CELL .
SINR optimization for MCS selection	Distributed massive MIMO cells ^a	32T/64T cells	None
Downlink rank robustness	Distributed massive MIMO cells ^a	32T/64T cells	None
Downlink small packet shaping	Distributed massive MIMO cells ^a	32T/64T cells	None
Power optimization for downlink retransmissions	None	32T/64T cells	None
Downlink retransmission scheduling priority adjustment	Distributed massive MIMO cells ^a	32T/64T cells	None

Function Name	Supported LampSite Cell Type	Supported Low-Frequency TDD Macro Cell Type	Additional Requirement
Downlink slot awareness scheduling optimization	Distributed massive MIMO cells ^a	32T/64T cells	None
Intelligent MCS optimization for downlink MU-MIMO	Distributed massive MIMO cells ^a	32T/64T cells	The NRDUCell.NrDuCellNetworking-Mode parameter is set to NORMAL_CELL .
Blind SRS interference suppression enhancement	None	32T/64T cells	The NRDUCell.NrDuCellNetworking-Mode parameter is set to NORMAL_CELL or FUSION_CELL .

a: Distributed massive MIMO LampSite cells are those with the **NRDUCell.LampSiteCellFlag** parameter set to **YES** and the **DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter selected.

5.3.5 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

[Table 5-2](#) describes the parameters used for activating iBeam 3.0.

Table 5-2 Parameters used for activating iBeam 3.0

Parameter Name	Parameter ID	Option	Setting Notes
High Precision Beam Phase3 Switch	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	None	Set this parameter to ON if iBeam 3.0 is required.

Table 5-3 describes the parameters used for activating weight enhancement.

Table 5-3 Parameters used for activating weight enhancement

Parameter Name	Parameter ID	Option	Setting Notes
Precoding Algorithm Switch	NRDUCellPdschPre-code.PrecodingAlgoSwitch	DL_SELF_FUSION_WEIGHT_SW	Select this option if PDSCH self-fusion weights are required.

Parameter Name	Parameter ID	Option	Setting Notes
Self Fusion Weight SRS PreSINR Thld	NRDUCellPdschPre-code.<i>SelfFusionWtSrsPresinrThld</i>	None	Retain the default value. This parameter takes effect only when the DL_SELF_FUSION_WEIGHT_SW option of the NRDUCellPdschPre-code.<i>PrecodingAlgoSwtch</i> parameter is selected. A larger value of this parameter results in a larger number of UEs for which PDSCH self-fusion weights take effect and a higher upper limit for the increase in the downlink user-perceived rate. A smaller value of this parameter results in the opposite effects. If the NRDUCellTrp.<i>TxRxMode</i> parameter is set to a value other than 4T4R, the NRDUCellPdschPre-code.<i>SelfFusionWtSrsPresinrThld</i> parameter must be set to a value in the range of -20 to 10.

Parameter Name	Parameter ID	Option	Setting Notes
Self Fusion Weight RHO Thld	NRDUCellPdschPre-code.<i>SelfFusionWtRh</i> <i>oThld</i>	None	Retain the default value. This parameter takes effect only when the DL_SELF_FUSION_WEIGHT_SW option of the NRDUCellPdschPre-code.<i>PrecodingAlgoSwitch</i> parameter is selected. A larger value of this parameter results in a larger number of UEs for which PDSCH self-fusion weights take effect and a larger increase in the downlink user-perceived rate. A smaller value of this parameter results in the opposite effects.
PDCCH Algorithm Switch	NRDUCellPdcch.<i>PdcchAlgoSwitch</i>	PDCCH_ROBUST_WEIGHT_SW	Select this option if PDCCH robust weights are required.

Parameter Name	Parameter ID	Option	Setting Notes
PDCCH Robust Weight SRS PreSINR Thld	NRDUCellPdcchAlgo.PdcchRobWtSrsPresinrThld	None	Retain the default value. This parameter takes effect only when the DL_SELF_FUSION_WEIGHT_SW option of the NRDUCellPdschPre-code.PrecodingAlgoSwitch parameter is selected. A smaller value of this parameter results in a higher proportion of times that PDCCH robust weights take effect and a larger increase in the downlink user-perceived rate. However, if this threshold is set to an excessively small value, the SRS measurement accuracy is reduced, and the downlink user-perceived rate decreases. A larger value of this parameter results in a lower proportion of times that PDCCH robust weights take effect and a smaller increase in the downlink user-perceived rate.

Parameter Name	Parameter ID	Option	Setting Notes
PDCCH Robust Weight BFGain Coefficient	NRDUCellPdcchAlgo.PdcchRobWtBFGainCoeff	None	Retain the default value. This parameter takes effect only when the DL_SELF_FUSION_WEIGHT_SW option of the NRDUCellPdschPre-code.PrecodingAlgoSwitch parameter is selected. A larger value of this parameter results in a larger beamforming gain of PDCCH robust weights, a more significant decrease in the average PDCCH aggregation level, and a larger increase in the downlink UE throughput. However, if the aggregation level is reduced excessively, the probability of missed detection on UEs increases, which may decrease the downlink UE throughput. A smaller value of this parameter results in a smaller beamforming gain of PDCCH robust weights and a smaller increase in the downlink UE throughput.

Table 5-4 describes the parameters used for activating scheduling enhancement.

Table 5-4 Parameters used for activating scheduling enhancement

Parameter Name	Parameter ID	Option	Setting Notes
DL MU-MIMO Scheduling Supplement Switch	NRDUCellDIMimo.DLMuMimoSchSupplementSw	DL_SMART_AMC_SW	Select this option if SINR optimization for MCS selection is required.
DL MCS Selection Algorithm Switch	NRDUCellDIAMc.DIMcsSelAlgoSw	ROBUST_RANK_SW	Select this option if downlink rank robustness is required.
CSI Resource Algo Switch	NRDUCellPucch.CsiResourceAlgoSwitch	CSI_REPORT_PERIOD_ADAPT_SWITCH	When downlink rank robustness is enabled, it is recommended that the following functions also be enabled for greater benefits: CSI reporting period adaptation (controlled by the CSI_REPORT_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.CsiResourceAlgoSwitch parameter) and PUCCH RB adaptation (controlled by the PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter).
PUCCH Algorithm Switch	NRDUCellPucch.PucchAlgoSwitch	PUCCH_RBRES_ADAPTIVE_SWITCH	
DL MCS Selection Algorithm Switch	NRDUCellDIAMc.DIMcsSelAlgoSw	ROBUST_RANK_MOBILITY_OPT_SW	For cells serving high-speed UEs, it is recommended that this option be selected to enable "rank robustness optimization in mobility scenarios".
Downlink Schedule Algorithm Switch	NRDUCellDISch.DISchAlgoSwitch	DL_SMALL_PACKET_SHAPE_SCH_SW	Select this option if downlink small packet shaping is required.

Parameter Name	Parameter ID	Option	Setting Notes
Channel Power Algorithm Switch	NRDUCellChnPwr.ChnPwrAlgoSwitch	DL_RETRANS_PWR_OPT_SW	Select this option if power optimization for downlink retransmissions is required.
Downlink Schedule Algorithm Switch	NRDUCellDISch.DIShAlgoSwitch	DL_RETRANS_PRI_ADJ_SW	Select this option if downlink retransmission scheduling priority adjustment is required.
Service Experience Algorithm Switch	NRDUCellServExp.ServiceExpAlgoSwitch	RES_BASED_MM_ADAPT_SCH_SW	When downlink retransmission scheduling priority adjustment is enabled, it is recommended that resource-based adaptive scheduling for massive MIMO also be enabled for greater benefits. (Resource-based adaptive scheduling for massive MIMO is controlled by the RES_BASED_MM_ADAPT_SCH_SW option of the NRDUCellServExp.ServiceExpAlgoSwitch parameter.)
Downlink Schedule Algorithm Switch	NRDUCellDISch.DIShAlgoSwitch	DL_SLOT_AWARE_SCH_OPT_SW	Select this option if downlink slot awareness scheduling optimization is required.
AI AMC Algorithm Switch	NRDUCellAiAlgo.AiAmcAlgoSwitch	DL_MU_MCS_INTEL_OPT_SW	Select this option if intelligent MCS optimization for downlink MU-MIMO is required.

Table 5-5 describes the parameters used for activating interference suppression enhancement.

Table 5-5 Parameters used for activating interference suppression enhancement

Parameter Name	Parameter ID	Option	Setting Notes
SRS Detection Algorithm Switch	NRDUCellSrs.SrsDetectionAlgoSwitch	SRS_ENH_B LIND_IS_SW	Select this option if blind SRS interference suppression enhancement is required.

5.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

To enable iBeam 3.0:

```
//Enabling iBeam 3.0
MOD NRDUCELLFEATURESW: NrDuCellId=0, HighPrecisionBeamPhase3Sw=ON;
```

To enable functions related to weight enhancement:

```
//Enabling PDSCH self-fusion weights
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, PrecodingAlgoSwitch=DL_SELF_FUSION_WEIGHT_SW-1;
//(Optional) Setting the SelfFusionWtSrsPresinrThld and SelfFusionWtRhoThld parameters for PDSCH self-fusion weights
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, SelfFusionWtSrsPresinrThld=2, SelfFusionWtRhoThld=80;
//(Optional, recommended when PDSCH self-fusion weights are used) Enabling downlink rank adaptation for cell edge UEs performing small-packet services
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRankSelAlgoSw=DL_SMALL_PKT_CEU_RANK_ADAPT_SW-1;
//Enabling PDCCH robust weights
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_ROBUST_WEIGHT_SW-1;
//(Optional) Setting the PdcchRobWtSrsPresinrThld and PdcchRobWtBFGainCoeff parameters for PDCCH robust weights
MOD NRDUCELLPDCCHALGO: NrDuCellId=0, PdcchRobWtSrsPresinrThld=-13, PdcchRobWtBFGainCoeff=6;
```

To enable functions related to scheduling enhancement:

```
//Enabling SINR optimization for MCS selection
MOD NRDUCELLDLMIMO: NrDuCellId=0, DLMuMimoSchSupplementSw=DL_SMART_AMC_SW-1;
//Enabling downlink rank robustness
MOD NRDUCELLDLAMC: NrDuCellId=0, DlMcsSelAlgoSw=ROBUST_RANK_SW-1;
//(Optional, recommended when downlink rank robustness is enabled) Enabling CSI reporting period adaptation and PUCCH RB adaptation for greater benefits
MOD NRDUCELLPUCCH: NrDuCellId=0, CsiResoureAlgoSwitch=CSI_REPORT_PERIOD_ADAPT_SWITCH-1, PucchAlgoSwitch=PUCCH_RBRES_ADAPTIVE_SWITCH-1;
//(Optional, recommended for cells serving high-speed UEs) Enabling "rank robustness optimization in mobility scenarios"
MOD NRDUCELLDLAMC: NrDuCellId=0, DlMcsSelAlgoSw=ROBUST_RANK_MOBILITY_OPT_SW-1;
```

```
//Enabling downlink small packet shaping
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=DL_SMALL_PACKET_SHAPE_SW-1;
//Enabling power optimization for downlink retransmissions
MOD NRDUCELLCHNPWR: NrDuCellId=0, ChnPwrAlgoSwitch=DL_RETRANS_PWR_OPT_SW-1;
//Enabling downlink retransmission scheduling priority adjustment
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=DL_RETRANS_PRI_ADJ_SW-1;
//((Optional, recommended when downlink retransmission scheduling priority adjustment is enabled)
Enabling resource-based adaptive scheduling for massive MIMO for greater benefits
MOD NRDUCELLSERVEXP: NrDuCellId=0, ServiceExpAlgoSwitch=RES_BASED_MM_ADAPT_SW-1;
//Enabling downlink slot awareness scheduling optimization
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=DL_SLOT_AWARE_SW-1;
//Enabling intelligent MCS optimization for downlink MU-MIMO
MOD NRDUCELLAIALGO: NrDuCellId=0, AiAmcAlgoSwitch=DL_MU_MCS_INTEL_OPT_SW-1;
```

To enable functions related to interference suppression enhancement:

```
//Enabling blind SRS interference suppression enhancement
MOD NRDUCELLSRS: NrDuCellId=0, SrsDetectionAlgoSwitch=SRS_ENH_BLIND_IS_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

To disable functions related to weight enhancement:

```
//Disabling PDCCH robust weights
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_ROBUST_WEIGHT_SW-0;
//((Optional) Disabling downlink rank adaptation for cell edge UEs performing small-packet services
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRankSelAlgoSw=DL_SMALL_PKT_CEU_RANK_ADAPT_SW-0;
//Disabling PDSCH self-fusion weights
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, PrecodingAlgoSwitch=DL_SELF_FUSION_WEIGHT_SW-0;
```

To disable functions related to scheduling enhancement:

```
//Disabling intelligent MCS optimization for downlink MU-MIMO
MOD NRDUCELLAIALGO: NrDuCellId=0, AiAmcAlgoSwitch=DL_MU_MCS_INTEL_OPT_SW-0;
//Disabling downlink slot awareness scheduling optimization
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=DL_SLOT_AWARE_SW-0;
//((Optional) Disabling resource-based adaptive scheduling for massive MIMO
MOD NRDUCELLSERVEXP: NrDuCellId=0, ServiceExpAlgoSwitch=RES_BASED_MM_ADAPT_SW-0;
//Disabling downlink retransmission scheduling priority adjustment
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=DL_RETRANS_PRI_ADJ_SW-0;
//Disabling power optimization for downlink retransmissions
MOD NRDUCELLCHNPWR: NrDuCellId=0, ChnPwrAlgoSwitch=DL_RETRANS_PWR_OPT_SW-0;
//Disabling downlink small packet shaping
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=DL_SMALL_PACKET_SHAPE_SW-0;
//((Optional) Disabling "rank robustness optimization in mobility scenarios"
MOD NRDUCELLDLAMC: NrDuCellId=0, DlMcsSelAlgoSw=ROBUST_RANK_MOBILITY_OPT_SW-0;
//((Optional) Disabling CSI reporting period adaptation and PUCCH RB adaptation
MOD NRDUCELLPUCCH: NrDuCellId=0, CsiResoureAlgoSwitch=CSI_REPORT_PERIOD_ADAPT_SWITCH-0,
PucchAlgoSwitch=PUCCH_RBRES_ADAPTIVE_SWITCH-0;
//Disabling downlink rank robustness
MOD NRDUCELLDLAMC: NrDuCellId=0, DlMcsSelAlgoSw=ROBUST_RANK_SW-0;
//Disabling SINR optimization for MCS selection
MOD NRDUCELLDLMIMO: NrDuCellId=0, DLMuMimoSchSupplementSw=DL_SMART_AMC_SW-0;
```

To disable functions related to interference suppression enhancement:

```
//Disabling blind SRS interference suppression enhancement
MOD NRDUCELLSRS: NrDuCellId=0, SrsDetectionAlgoSwitch=SRS_ENH_BLIND_IS_SW-0;
```

To disable iBeam 3.0:

```
//Disabling iBeam 3.0
MOD NRDUCELLFEATURESW: NrDuCellId=0, HighPrecisionBeamPhase3Sw=OFF;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Observe the **User Downlink Average Throughput (DU)** before and after iBeam 3.0 is enabled. If the value of this indicator increases, iBeam 3.0 has taken effect.

5.4.3 Network Monitoring

After iBeam 3.0 is enabled, use the performance indicators described in [5.2 Network Analysis](#) to monitor cell performance.

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

UE Power Saving

Massive MIMO AHR (Low-Frequency TDD)

RedCap

Downlink Scheduling

URLLC

Channel Management

High Speed Mobility

MIMO (TDD)

Interference Avoidance

LTE and NR Power Allocation

NR Inter-Carrier Dynamic Power Allocation

3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"

3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"

Energy Conservation and Emission Reduction

Beam Management

Coordinated Interference Management (Low-Frequency TDD)

Distributed Antenna Solutions (Low-Frequency TDD)

License Control Item Lists

Co-MM (Low-Frequency TDD)

Standards Compliance

FWA

Coordinated Interference Management (Low-Frequency TDD)

Multi-Frequency Smart Aggregation

Super Uplink

Positioning

Distributed Antenna Solutions (Low-Frequency TDD)

Indoor Coverage

LTE and NR Zero Bufferzone

Interference Avoidance

CoMP

Network Slicing

DRX

ANR

Network Slicing

Cell Combination

CA Experience Improvement